

Evidence Appendix

MultiPlanetary Exchange System · Milestone 2 · Design v3

Introduction

All results come from one discrete-event simulator that implements the brief’s network in full (moving-receiver light time, solar exclusion, links, queues, timers, sessions, quotas, maintenance, incidents and random loss) and the ledgers of design v3. In every traced run it checks conservation, non-negative balances, hold arithmetic and registry custody after every ledger event.

A trace is a single run *without packet loss* unless it is labelled “lossy”. Times are in hours after the start of the scenario, and the epoch is 2026-09-22 00:00 TDB. For every probability we state whether it is exact or an approximation.

E1. Orbital calculations

Method. Each body follows the fixed Kepler ellipse given in `data.zip`. The mean anomaly is $M = M_0 + nt$, where n is the supplied mean motion. We solve Kepler’s equation by Newton iteration until the residual is below 10^{-15} rad, and then rotate by ω , i and Ω into the heliocentric J2000 ecliptic frame. Units are AU and seconds of TDB since the epoch; tables give times in hours. The relays use the same code with $e = 0$ and the supplied phases. The largest difference from the brief’s table of epoch positions is 7.3e-07 AU, which is the rounding of the table (Table A1). Table A2 gives later positions. As the brief asks, every body and relay returns to its starting point after one orbital period to within 3e-14 AU, and over 200,001 samples per orbit its distance from the Sun stays within $[a(1 - e), a(1 + e)]$ to within 4e-16 AU, the floating-point rounding.

Light time. A packet emitted at time t_e from body P reaches body Q at the time t_a that solves $t_a = t_e + 499.02 |Q(t_a) - P(t_e)|$, where 499.02 is the light time in seconds per AU. We solve this by fixed-point iteration. For the Earth to Relay A link the successive corrections are 2.1e-02, 4.1e-07, 7.7e-12 seconds. Each step is smaller than the previous one by a factor of about $v/c \approx 10^{-4}$, so two iterations reach 1 ms. We always iterate to 10^{-6} s.

Solar exclusion. A launch is blocked if the line segment from $P(t_e)$ to $Q(t_a)$ passes within 0.10 AU of the Sun. Every launch, ACK and receipt is checked at its own emission time.

Table A1: Positions at the epoch and their difference from the table in the brief.

Body	x (AU)	y (AU)	z (AU)	Error (AU)
Mercury	-0.213426	-0.410281	-0.013955	1.8e-07
Venus	+0.679067	-0.257950	-0.042725	3.4e-07
Earth	+1.003581	-0.023693	-0.000003	5.3e-07
Mars	+0.247476	+1.526869	+0.025929	7.3e-07
Ceres	+0.375786	+2.653941	+0.014791	5.6e-07
Jupiter	-3.438179	+4.038309	+0.060149	5.2e-07
Saturn	+9.271221	+1.717485	-0.398962	3.8e-07
Uranus	+8.962475	+17.253919	-0.052129	6.0e-07
Neptune	+29.839098	+1.351785	-0.715470	3.3e-07
Relay A	+2.000000	+2.000000	+0.000000	4.4e-16
Relay B	-2.000000	+2.000000	+0.000000	4.4e-16

Effect of receiver motion. Table A3 shows the S1a funding route (Earth to Relay A to Mars) and the return route for the shares (Mars to Relay A to Earth) at their actual emission times in the S1a run: hour 1.1317 for the funding, after the session handshake, and hour 2.5027 for the shares, just after the fill. Each relay sends the packet on 2 s after it arrives (1 s of processing and 1 s of serialization). The “frozen” column uses the positions of both ends at the emission time. The forward and return legs are not mirror images. On the Earth to Relay A leg the flight is +21.4 ms longer than the frozen value because Relay A moves away during the flight.

Timers. T_0 is the route time with empty queues. The endpoint timer is $R_e = 2T_0 + 24$ h and the hop timer is $R_h = 2 \times \text{flight} + 60$ min. On the S1 routes: Ea-A-Ma at hour 0.0000: $T_0 = 33.91$ min, $R_e = 25.13$ h,

Table A2: Positions computed by the same code at hour 300 and after one Julian year, for the bodies used in S1.

Body	Hour 300 (x, y, z) in AU	Hour 8,766 (x, y, z) in AU
Earth	(+0.9822, +0.1894, -0.0000)	(+1.0036, -0.0192, -0.0000)
Mars	(+0.0802, +1.5602, +0.0307)	(-0.4854, -1.4173, -0.0178)
Ceres	(+0.2446, +2.6592, +0.0391)	(-2.4828, +0.3753, +0.4693)
Relay A	(+1.9076, +2.0883, +0.0000)	(-1.4432, +2.4326, +0.0000)
Relay B	(-2.0883, +1.9076, +0.0000)	(-2.4326, -1.4432, -0.0000)
Neptune	(+29.8370, +1.3912, -0.7162)	(+29.7556, +2.5017, -0.7374)

Table A3: Flight times on the S1a routes with a moving receiver, compared with the “frozen” time that uses both positions at emission. The first four rows are the S1a funding and share delivery at their actual launch times; the last four are illustrative emissions at the epoch.

Path	Emitted (s)	Flight (s)	Frozen (s)	Difference (ms)	Distance (AU)	Clearance (AU)
Earth→A	4074.049	1125.375451	1125.354049	+21.402	2.2552	1.004
A→Mars	5201.424	906.214978	906.152494	+62.484	1.8160	1.547
Mars→A	9009.541	906.276241	906.302932	-26.691	1.8161	1.547
A→Earth	9917.818	1124.844810	1124.944916	-100.106	2.2541	1.004
Earth→Mars	2.000	860.995877	860.951580	+44.297	1.7254	0.892
Mars→Earth	2.000	860.875291	860.951580	-76.289	1.7251	0.892
Neptune→A	1.000	13901.212047	13900.618008	+594.039	27.8570	2.828
A→Neptune	1.000	13900.598751	13900.618008	-19.257	27.8558	2.828

$R_h = 97.5$ and 90.2 min; Ma-A-Ea at hour 1.6966: $T_0 = 33.90$ min, $R_e = 25.13$ h, $R_h = 90.2$ and 97.5 min; Ce-A-Ma at hour 0.0000: $T_0 = 29.71$ min, $R_e = 24.99$ h, $R_h = 89.1$ and 90.2 min; Ne-A-Ma at hour 0.0000: $T_0 = 246.85$ min, $R_e = 32.23$ h, $R_h = 523.4$ and 90.2 min.

F. Message formats and sizes

Every message has a fixed-width binary layout. The tables below give every field, and the record sizes are the ones the simulator uses for packing. Integers are unsigned big-endian unless a field says otherwise. Times are signed 64-bit integer milliseconds of TDB since the epoch. Money is in integer cents, and share quantities are whole shares. Each record carries the signature of the principal that issued it, so it can be checked at any exchange whatever session or forwarder carried it.

Table A4: Common header of every application record.

Field	Bytes and meaning
Type	1: Record type code
Version	1: Format version (1)
Length	2: Record length in bytes
Record ID	12: 32-bit principal prefix and 64-bit counter; never reused
Created	8: Signed 64-bit integer milliseconds of TDB since the epoch
Signature	64: Ed25519 by the issuing principal over every other byte of the record
Total	88

Packing and splitting rules. (1) A record is never split across packets. The sender takes records from the outbox for one peer in order and adds whole records while the payload stays within 960 bytes (16-byte sub-header plus at most 944 bytes of records); the next record starts a new packet. Every fixed record is at most 256 bytes, so up to 5 transfers, 7 receipts or 3 contract instructions fit in one packet (Table A7). The simulator’s packer asserts that every record is between 1 and 944 bytes. (2) STATUS is the only variable-length record. It is closed at 52 entries (944 bytes). If more remote clients need an entry at the same moment, the host writes further STATUS records, each a complete signed record of its own. (3) A client may batch whole instructions in one direct packet, up to 944 bytes. A larger batch is split into several direct packets, and each one counts against the client’s 12 per day. A batch that may need forwarding (R8) is limited to

Table A5: Fixed-width body of each record type. The totals are the sizes the simulator charges (`protocol.SIZE`); the build checks that the fields add up to them. A STATUS record grows by 16 bytes for each entry after the first.

Record	Bytes	Fields after the common header (bytes)
TRANSFER	176	Source exchange 1; Destination exchange 1; Purpose 1; Asset 1; Owner account 8; Amount (units or cents) 8; Number n to this destination 8; Reference ID (order, contract, request) 12; Beneficiary account 8; Terms digest (SHA-256) 32; Reserved 8
RECEIVED	120	Transfer ID 12; Source exchange 1; Result (credited, restricted) 1; Import time 8; Watermark W for this source 8; Reserved 2
ORDER	128	Account 8; Host 1; Asset 1; Side 1; Time in force 1; Quantity 8; Limit price (cents) 8; Expiry 8; Disposition (keep, home) 1; Reserved 3
CANCEL	128	Account 8; Target order ID 12; Reserved 20
WITHDRAW	128	Account 8; Asset 1; Destination exchange 1; Amount 8; Reserved 22
CLOSE	128	Account 8; Reserved 32
CINSTR	256	Account 8; Contract ID 12; Role (long, short) 1; Quantity Q 8; Multiplier 4; Opening price K 8; Floor 8; Cap 8; Five observation times 40; Maturity 8; Fixing time 8; Opening deadline 8; Counterparty account 8; Price source 8; Margin 8; Reserved 23
STATUS	128	Host exchange 1; Home exchange 1; Entry count m (1 to 52) 2; As-of time 8; Reserved 12; Entry 1: account 8, contract counter 6, state 1, flags 1 16
PRICE	200	Contract ID 12; Observation number 1; Index value (fixed point) 8; Observation time 8; Price source 8; Source attestation (Ed25519) 64; Reserved 11
MDS	152	Market exchange 1; Asset 1; Book version 8; Best bid (cents) 8; Bid size 4; Best ask (cents) 8; Ask size 4; Last trade price (cents) 8; Last trade size 4; Last trade time 8; Reserved 10
MDREQ	120	Requesting exchange 1; Asset 1; Reserved 30

Table A6: Packet layouts. Every packet is at most 1,024 bytes: the brief's 64-byte header and at most 960 bytes of payload.

Unit	Bytes	Layout
Network header (brief)	64	Every packet; defined by the network
Data sub-header	16	Session 8, data sequence 4, record count 2, flags 2
Data packet	64 + 16 + records	Whole records, at most 944 bytes of records
SYN, SYN-ACK, ACK, DACK payload	96	Session 8, route 4, data sequence 4, attempt 1, kind 1, reserved 6, time 8, signature 64
Hop receipt payload	16	Packet ID 12, launch number 1, reserved 3
Direct sub-header	16	Principal 8, packet sequence 4, instruction count 1, flags 1, reserved 2
Direct packet	64 + 16 + instructions	Whole signed instructions, at most 944 bytes
FWD wrapper (R8)	24 + inner	Forwarder 1, target host 1, attempt 1, flags 1, forward time 8, origin principal 8, inner length 2, reserved 2

920 bytes so that the 24-byte FWD header still fits; a forwarding exchange forwards each received packet as one FWD packet and never merges or splits them. The simulator asserts this limit on every direct send. (4) RECEIVED records are not sent alone if other records for the same peer are waiting: they share packets with whatever is in the outbox. A packet that holds only receipts uses the reserve quota (Section E6).

Examples. The S1a funding packet holds one TRANSFER: $64 + 16 + 176 = 256$ bytes. A backlog packet with five transfers is $64 + 16 + 880 = 960$ bytes. A STATUS record with one entry is 128 bytes and one

Table A7: Capacity. A data or direct packet carries a 16-byte sub-header and up to 944 bytes of whole records. “Packet bytes” is the full packet at that count, header included; it never exceeds 1,024. The last column is the number of client instructions that still fit when R8 wraps the batch in a 24-byte FWD header.

Record	Bytes	Per packet	Packet bytes	Per FWD packet
TRANSFER	176	5	960	—
RECEIVED	120	7	920	—
ORDER	128	7	976	7
CANCEL	128	7	976	7
WITHDRAW	128	7	976	7
CLOSE	128	7	976	7
CINSTR	256	3	848	3
PRICE	200	4	880	—
MDS	152	6	992	—
MDREQ	120	7	920	—
STATUS (52 entries)	944	1	1024	—

with 52 entries is 944 bytes. The S1i batch of two orders is a direct packet of $64 + 16 + 256 = 336$ bytes, and a forwarded order is $64 + 16 + 24 + 128 = 232$ bytes.

S1. Scenario traces

These runs have no packet loss, start at the epoch and include the scheduled maintenance. Each scenario is a separate run from the opening balances. A scenario is complete when every intended result is spendable at its owner’s home. Packet counts include every backbone launch (data, handshakes, hop receipts, ACKs and retries) and every direct copy.

Transactions and value settled. Both efficiency measures use one definition. A *transaction* is a completed transfer (funding, home delivery, payout or withdrawal), a trade fill, or a settled futures contract. *Value settled* adds their values as the brief defines them: a fill counts price $\times$ quantity, a transfer counts the cash moved or the shares at the stated \$100, and a contract counts its notional $Q \times 1 \times 100$. S1a has three transactions (funding \$100,000, the fill \$100,000 and the delivery of the shares \$100,000), so it settles \$300,000; S1d has five (two fundings, the \$200,000 notional and two payouts). *Encumbered* value is the sum of order holds, contract margin and value in transit, given per asset as the peak value and the asset-hours. *Utilisation* is the peak encumbered cash divided by the total cash of \$500,000. *Exposure* is a side’s largest possible loss to its counterparty. It is \$0 for equities, because cash and shares change owner in one ledger event at the host (the buyer carries only the market risk of the shares it chose to own). For futures it is the cap $40Q$, fully escrowed before the contract opens. Tables A8 and A9 give the results.

Table A8: Completion and every packet of the S1 runs without loss, by type: session handshakes (SYN, SYN-ACK, ACK), record launches (DATA), endpoint acknowledgements (DACK), hop receipts (RCPT) and direct copies. Tx counts completed transactions as defined in the text.

	Done (h)	SYN	S-ACK	ACK	DATA	DACK	RCPT	Backbone	Direct	Total	Tx	Pkts/tx
S1a	3.07	2	2	2	8	8	22	44	3	47	3	15.7
S1b	3.16	2	2	2	8	8	22	44	3	47	3	15.7
S1c	15.82	2	2	2	8	8	22	44	8	52	3	17.3
S1d	328.17	8	8	8	22	22	68	136	5	141	5	28.2
S1e	328.17	8	8	8	22	22	68	136	5	141	5	28.2
S1f	328.17	8	8	8	22	22	68	136	5	141	5	28.2
S1g	—	2	2	2	12	12	30	60	6	66	2	33.0
S1h	—	0	0	0	0	0	0	0	0	0	0	—
S1i	51.06	2	2	2	8	8	22	44	15	59	7	8.4

Per customer. The brief’s measures count every leg, so a \$100,000 purchase settles \$300,000 and costs 15.7 packets per transaction. Table A10 gives the same runs from the client’s side: one purchase with home delivery costs 47 packets and commits exactly its own price, and the two futures sides together post 80% of the notional (40% each). Funding a remote market moves the client’s own money to its own account

Table A9: Capital measures (Section 7 of the brief). Asset-hours are in thousands (k). Capital efficiency is peak encumbered cash divided by value settled. Exp./side is each side’s largest possible loss; S1g and S1h never open.

Scenario	Value settled	Peak enc. / asset-h	Util.	Cap. eff.	Exp./side
S1a Earth buys Ares	\$300,000	\$100k / 169.6k; 1,000 Ares / 3.1k	20%	0.33	\$0
S1b Earth buys Belt	\$300,000	\$100k / 166.9k; 1,000 Belt / 3.2k	20%	0.33	\$0
S1c Uranus buys Terra	\$120,000	\$40k / 316.5k; 400 Terra / 6.3k	8%	0.33	\$0
S1d rising	\$520,000	\$160k / 52,386.7k	32%	0.31	\$80k
S1e falling	\$520,000	\$160k / 52,372.5k	32%	0.31	\$80k
S1f $Q = 2,500$	\$650,000	\$200k / 65,483.4k	40%	0.31	\$100k
S1g $Q = 2,501$	\$200,080	\$100k / 16,782.1k	20%	0.50	\$0
S1h $Q = 4,000$	\$0	—	0%	—	\$0
S1i Repeat trading	\$23,020	\$10k / 106k; 15 Ares / 51	2%	0.43	\$0

there; the purchase is complete only when the shares are spendable at home, as the brief defines completion, and nothing remains encumbered afterwards. The funding leg is counted as a transaction because the brief counts completed transfers, but it is not a completed purchase. The component capital efficiency of 0.33 in Table A9 therefore reflects how the workflow is divided into legs; it does not mean that a client needs only a third of the price. Every purchase is fully prefunded.

Table A10: Efficiency per customer action (no loss). Value is the purchase price, or the notional for futures (both sides). Committed / value is the peak encumbered cash over that value; hours committed are dollar-hours over the peak.

Customer action	Value	Packets	Committed / value	Hours committed	Done (h)
Alice buys 1,000 Ares (Earth at Mars)	\$100,000	47	1.00	1.7	3.1
Alice buys 1,000 Belt (Earth at Ceres)	\$100,000	47	1.00	1.7	3.2
Eve buys 400 Terra (Uranus at Earth)	\$40,000	52	1.00	7.9	15.8
Futures, $Q = 2,000$, both sides	\$200,000	141	0.80	327.4	328.2
Repeat trading, \$10,000 funded	\$10,000	59	1.00	10.6	51.1

Scale and the binding constraint. The futures in S1d and S1e hold \$160,000, or 32% of all cash, in escrow for 324 hours. The boundary case $Q = 2,500$ holds \$200,000, or 40%. With $Q = 2,501$ each side needs \$100,040. Cara has only \$100,000, so her Ceres exchange rejects her funding transfer immediately. Alice’s \$100,040 is held at Mars until the 168-hour deadline, then released. Mars reports REJECTED to Earth in a STATUS record (R5), which reaches Alice at hour 168.56, and she withdraws the cash home by hour 169.79. With $Q = 4,000$ (\$160,000 each) both home exchanges reject the funding before anything moves. All 91 ledger checks in these runs passed.

Traces. Tables A11 and A12 give the financial traces of S1a and S1d. They omit rows that change no balance: an exchange learning that its transfer arrived, and Bob publishing each observation. Section T gives the complete trace of every S1 run, S1a to S1i, including those rows. The packet-level communication traces are machine-readable files in `milestone2/traces/`, one set per S1 run and per S3 relocation. `*_launches.csv` lists every packet in Table A8, one row each, with its type, session, copy number, emission and arrival time, distance, loss probability and outcome. `*_transport.csv` lists every session and recovery event, and `*_ledger.csv` every ledger event. The packet counts in Table A8 are computed from these files.

Table A11: Full trace of S1a: Alice (Earth) buys 1,000 Ares from Bob (Mars) at \$100.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Earth exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ea→Ma, batched with queued records	1.6966	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Earth balance	Funds own Mars account: \$100k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$50k at Earth, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.6966	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Ea, batched with queued records	2.2622	Alice \$50k at Earth, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2622	Earth exchange	MDS from Mars	Posts Ares snapshot to its board (0.57 h old); answers waiting QUOTES	Local access (1 s)	2.2625	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2625	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 0.57 h old	—	—	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2625	Alice	Published geometry; own funding confirmed	Sends ORDER ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5024	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Earth, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.5024	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Earth, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home
2.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Earth	Session from h 0.0006; DATA Ma→A h 2.5027, A→Ea h 2.7549	3.0674	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.5190	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.5357	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.0674	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	3.6331	Alice \$50k at Earth, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

Table A12: Full trace of S1d: capped futures on the rising path (Alice long, Cara short, $Q = 2,000$).

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$80k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$80k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$80k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Cara \$80k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k in transit

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Cara \$80k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k free at host
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$80k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
1.9829	Cara	Published geometry; own funding confirmed	Sends CINSTR $\times 2$ copies	Direct Ce $\rightarrow$ Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
2.1410	Mars exchange	Instruction + own funded balance	Holds Cara margin \$80k	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.1410	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma $\rightarrow$ Ce	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.1576	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR $\times 3$ copies	Direct Ea $\rightarrow$ Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$80k	—	—	Alice \$70k at Earth, \$80k held; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Both margins held	Opens F1: escrow \$160k; maturity h 302.5024, fixing h 326.5024	STATUS (OPEN) records to Earth and Ceres (backbone)	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma $\rightarrow$ Ea	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma $\rightarrow$ Ce	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5190	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5357	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.6363	Ceres exchange	STATUS from Mars: F1 HELD for Cara	Tells Cara (no balance change)	Local access (1 s)	2.6365	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.9980	Ceres exchange	STATUS from Mars: F1 OPEN for Cara	Tells Cara (no balance change)	Local access (1 s)	2.9982	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
3.0674	Earth exchange	STATUS from Mars: F1 OPEN for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
62.5 to 302.5	Mars exchange	Signed observation from Bob	Records observations 110, 120, 130, 135, 125 (every 60 h)	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
326.5024	Mars exchange	Fixing cutoff; uses observation 5	Fixes at 125: long net \$50k; gross long \$130k, short \$30k	—	settles; claims backed	Alice \$70k at Earth, \$130k free at host; Cara \$20k at Ceres, \$30k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$130k; TRANSFER MA-X1 (payout) to Earth	New session (SYN h 326.5027); DATA Ma→A h 327.6171, A→Ea h 327.8825	328.1736	Alice \$70k at Earth, \$130k in transit; Cara \$20k at Ceres, \$30k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Cara \$30k; TRANSFER MA-X2 (payout) to Ceres	New session (SYN h 326.5029); DATA Ma→A h 327.5228, A→Ce h 327.7882	328.0321	Alice \$70k at Earth, \$130k in transit; Cara \$20k at Ceres, \$30k in transit
328.0321	Ceres exchange	Authentic new transfer MA-X2	Imports once; credits Cara \$30k (spendable now)	RECEIVED to Mars (backbone)	328.5422	Alice \$70k at Earth, \$130k in transit; Cara \$50k at Ceres
328.1736	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$130k (spendable now)	RECEIVED to Mars (backbone)	328.7309	Alice \$200k at Earth; Cara \$50k at Ceres

S1e (falling prices). This run is identical to S1d, step by step, until the first observation, as the margin rule requires. The contract opens at the same time (2.5024 h) with the same packets. The observations are 90, 80, 70, 65 and 75. The fixing at 75 gives the long a net result of \$-50,000, so the gross payouts are reversed: \$30,000 to Alice and \$130,000 to Cara. Alice ends with \$100,000 and Cara with \$150,000 at

home. Cara’s payout is spendable at 328.032 h and Alice’s at 328.174 h. Table A13 gives the key steps of the remaining scenarios.

Table A13: Key steps of the scenarios without a trace table here (complete traces of every run: Section T).

Key steps (hours)	
S1d	Alice and Cara each fund \$80,000 at Mars (receipts known at 2.262 and 1.983) and send their contract instructions. Both margins are held and the contract opens at 2.502 with \$160,000 in escrow; Cara learns OPEN at 2.998 and Alice at 3.068. Bob publishes 110, 120, 130, 135 and 125 every 60 h; maturity at 302.502, fixing at 326.502 at 125 (long net +\$50,000). Payouts: \$30,000 spendable by Cara at 328.032 and \$130,000 by Alice at 328.174.
S1b	funds at Ceres at 1.670, receipt known at 2.226, fill at 2.608, Cara can spend at 2.608, Belt shares home at 3.164. Same structure as S1a on the route Earth, Relay A, Ceres. Mars is not involved.
S1c	funds at Earth at 7.913, receipt at Uranus at 10.551, fill at 13.188, Fin can spend at 13.188, Terra shares at Uranus at 15.824. Eve sends 8 copies from Uranus to Earth, each lost with probability 0.78.
S1f	As S1d with \$100,000 margin each. The contract opens at 2.502 and the payouts are spendable at 328.03 and 328.17. Alice ends with \$212,500 and Cara with \$37,500.
S1i	Alice funds \$10,000 at Mars. One batched direct packet carries two buy orders, each 40 at \$100 (o1 and o2). o1 fills 10 at \$97, and the \$30 price improvement is refunded immediately, then 5 at \$100. o2 is cancelled unfilled at +6 h, which releases its \$4,000 hold. At +12 h she sells 10 at \$101 to Bob. At +24 h her cancel of o1 races an immediate-or-cancel sell from Bob. Bob’s order arrives first and fills 5 at \$100, and the cancel then releases only the remainder, \$2,000 for 20 shares. Finally, her cash and shares are sent home. Done at 51.06 h, with 59 packets for 7 transactions.

T. Complete traces

These tables give every row of every S1 run and of the three S3 relocations discussed in the text, in the brief’s seven-column format. Rows come from the simulator’s event stream. The financial state is the ledger snapshot immediately after the event, for the parties named in the caption: “free at host” is available at the market, “held” is an order or instruction hold, and “margin” is contract escrow. A local instruction is shown when it arrives at the exchange, 1 s after the client sends it (the send time is in the packet column), so that every row’s state is the state after that row. Packet columns list each backbone DATA launch of the record by hop; the handshake, acknowledgements and hop receipts of every session are in `traces/<run>_launches.csv`, whose counts are Table A8. The runs have no loss, so every first launch arrives.

Table A14: Complete trace of S1a: Alice (Earth) buys 1,000 Ares from Bob (Mars) at \$100. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Earth exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ea→Ma, batched with queued records	1.6966	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Earth balance	Funds own Mars account: \$100k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$50k at Earth, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.6966	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Ea, batched with queued records	2.2622	Alice \$50k at Earth, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2622	Earth exchange	MDS from Mars	Posts Ares snapshot to its board (0.57 h old); answers waiting QUOTES	Local access (1 s)	2.2625	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2625	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 0.57 h old	—	—	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.2625	Alice	Published geometry; own funding confirmed	Sends ORDER ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$50k at Earth, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.5024	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Earth, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.5024	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Earth, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home
2.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Earth	Session from h 0.0006; DATA Ma→A h 2.5027, A→Ea h 2.7549	3.0674	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.5190	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.5357	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.0674	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	3.6331	Alice \$50k at Earth, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home
3.6331	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

Table A15: Complete trace of S1b: Alice (Earth) buys 1,000 Belt from Cara (Ceres) at \$100. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Cara	Own inventory at the market	Sell 1,000 Belt limit \$100 GTC	Local access at Ceres, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt home
0.0003	Ceres exchange	Its own available balances	Admits Cara EQB-sell; holds 1,000 Belt	—	—	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt held
0.0003	Alice	Listing table: Belt trades at Ceres	QUOTE for Belt	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt held
0.0003	Earth exchange	No fresh Belt snapshot on its board	Quote request (MDREQ) for Belt to Ceres	Backbone Ea→Ce, batched with queued records	1.6697	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt held
0.0003	Alice	Own Earth balance	Funds own Ceres account: \$100k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres, 1,000 Belt held
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER EA-X1 (funding) to Ceres	New session (SYN h 0.0006); DATA Ea→A h 1.1138, A→Ce h 1.4269	1.6697	Alice \$50k at Earth, \$100k in transit; Cara \$100k at Ceres, 1,000 Belt held
1.6697	Ceres exchange	Its own live book	Replies with Belt snapshot (MDS, version 1)	Backbone Ce→Ea, batched with queued records	2.2263	Alice \$50k at Earth, \$100k in transit; Cara \$100k at Ceres, 1,000 Belt held
1.6697	Ceres exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Earth (backbone)	2.2263	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.2263	Earth exchange	MDS from Ceres	Posts Belt snapshot to its board (0.56 h old); answers waiting QUOTES	Local access (1 s)	2.2266	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.2263	Earth exchange	Ceres imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.2266	Alice	Home board (local access)	Reads Belt quote: bid none, ask \$100 × 1,000, 0.56 h old	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held
2.2266	Alice	Published geometry; own funding confirmed	Sends ORDER ×3 copies	Direct Ea→Ce, launches 2.2266, 2.2432, 2.2599; p(loss) 0.197 each, all lost 7.7e-03	2.6085 (first)	Alice \$50k at Earth, \$100k free at host; Cara \$100k at Ceres, 1,000 Belt held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.6085	Ceres exchange	Its own available balances	Admits Alice EQB-buy; holds \$100k	—	—	Alice \$50k at Earth, \$100k held; Cara \$100k at Ceres, 1,000 Belt held
2.6085	Ceres exchange	Both holds in its own ledger	Fill 1,000 Belt @ \$100: Alice buys from Cara	—	local, atomic	Alice \$50k at Earth, 1,000 Belt at market; Cara \$200k at Ceres
2.6085	Ceres exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Belt; TRANSFER CE-X1 (automatic home delivery) to Earth	Session from h 0.0006; DATA Ce→A h 2.6087, A→Ea h 2.8520	3.1644	Alice \$50k at Earth, 1,000 Belt in transit; Cara \$200k at Ceres
2.6251	Ceres exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Belt in transit; Cara \$200k at Ceres
2.6418	Ceres exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, 1,000 Belt in transit; Cara \$200k at Ceres
3.1644	Earth exchange	Authentic new transfer CE-X1	Imports once; credits Alice 1,000 Belt (spendable now)	RECEIVED to Ceres (backbone)	3.7210	Alice \$50k at Earth, 1,000 Belt home; Cara \$200k at Ceres
3.7210	Ceres exchange	Earth imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, 1,000 Belt home; Cara \$200k at Ceres

Table A16: Complete trace of S1c: Eve (Uranus) buys 400 Terra from Fin (Earth) at \$100. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Fin	Own inventory at the market	Sell 400 Terra limit \$100 GTC	Local access at Earth, sent h 0.0000	0.0003	Eve \$50k at Uranus; Fin \$50k at Earth, 1,000 Terra home
0.0003	Earth exchange	Its own available balances	Admits Fin EQT-sell; holds 400 Terra	—	—	Eve \$50k at Uranus; Fin \$50k at Earth, 600 Terra home, 400 Terra held
0.0003	Eve	Listing table: Terra trades at Earth	QUOTE for Terra	Local access at Uranus, sent h 0.0000	0.0003	Eve \$50k at Uranus; Fin \$50k at Earth, 600 Terra home, 400 Terra held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Uranus exchange	No fresh Terra snapshot on its board	Quote request (MDREQ) for Terra to Earth	Backbone Ur→Ea, batched with queued records	7.9134	Eve \$50k at Uranus; Fin \$50k at Earth, 600 Terra home, 400 Terra held
0.0003	Eve	Own Uranus balance	Funds own Earth account: \$40k	Local access at Uranus, sent h 0.0000	0.0003	Eve \$50k at Uranus; Fin \$50k at Earth, 600 Terra home, 400 Terra held
0.0003	Uranus exchange	Instruction or fill; its own ledger	Debits Eve \$40k; TRANSFER UR-X1 (funding) to Earth	New session (SYN h 0.0006); DATA Ur→A h 5.2765, A→Ea h 7.6012	7.9134	Eve \$10k at Uranus, \$40k in transit; Fin \$50k at Earth, 600 Terra home, 400 Terra held
7.9134	Earth exchange	Its own live book	Replies with Terra snapshot (MDS, version 1)	Backbone Ea→Ur, batched with queued records	10.5510	Eve \$10k at Uranus, \$40k in transit; Fin \$50k at Earth, 600 Terra home, 400 Terra held
7.9134	Earth exchange	Authentic new transfer UR-X1	Imports once; credits Eve \$40k (spendable now)	RECEIVED to Uranus (backbone)	10.5510	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
10.5510	Uranus exchange	MDS from Earth	Posts Terra snapshot to its board (2.64 h old); answers waiting QUOTES	Local access (1 s)	10.5513	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
10.5510	Uranus exchange	Earth imported UR-X1	Marks transfer delivered; no balance change	—	—	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
10.5513	Eve	Home board (local access)	Reads Terra quote: bid none, ask \$100 × 400, 2.64 h old	—	—	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
10.5513	Eve	Published geometry; own funding confirmed	Sends ORDER ×8 copies	Direct Ur→Ea, launches 10.5513, 10.5680, 10.5846, 10.6013, 10.6180, 10.6346, 10.6513, 10.6680; p(loss) 0.781 each, all lost 1.4e-01	13.1878 (first)	Eve \$10k at Uranus, \$40k free at host; Fin \$50k at Earth, 600 Terra home, 400 Terra held
13.1878	Earth exchange	Its own available balances	Admits Eve EQT-buy; holds \$40k	—	—	Eve \$10k at Uranus, \$40k held; Fin \$50k at Earth, 600 Terra home, 400 Terra held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
13.1878	Earth exchange	Both holds in its own ledger	Fill 400 Terra @ \$100: Eve buys from Fin	—	local, atomic	Eve \$10k at Uranus, 400 Terra at market; Fin \$90k at Earth, 600 Terra home
13.1878	Earth exchange	Instruction or fill; its own ledger	Debits Eve 400 Terra; TRANSFER EA-X1 (automatic home delivery) to Uranus	Session from h 0.0006; DATA Ea→A h 13.1881, A→Ur h 13.5004	15.8244	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2045	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2211	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2378	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2545	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2711	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.2878	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
13.3045	Earth exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Eve \$10k at Uranus, 400 Terra in transit; Fin \$90k at Earth, 600 Terra home
15.8244	Uranus exchange	Authentic new transfer EA-X1	Imports once; credits Eve 400 Terra (spendable now)	RECEIVED to Earth (backbone)	18.4611	Eve \$10k at Uranus, 400 Terra home; Fin \$90k at Earth, 600 Terra home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
18.4611	Earth exchange	Uranus imported EA-X1	Marks transfer delivered; no balance change	—	—	Eve \$10k at Uranus, 400 Terra home; Fin \$90k at Earth, 600 Terra home

Table A17: Complete trace of S1d: capped futures, rising path, $Q = 2,000$ (Alice long, Cara short). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$80k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$80k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$80k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Cara \$80k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k in transit
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Cara \$80k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k free at host
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$80k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
1.9826	Ceres exchange	Mars imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
1.9829	Cara	Published geometry; own funding confirmed	Sends CINSTR ×2 copies	Direct Ce→Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
2.1410	Mars exchange	Instruction + own funded balance	Holds Cara margin \$80k	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.1410	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma→Ce	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.1576	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$80k	—	—	Alice \$70k at Earth, \$80k held; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Both margins held	Opens F1: escrow \$160k; maturity h 302.5024, fixing h 326.5024	STATUS (OPEN) records to Earth and Ceres (backbone)	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma→Ea	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma→Ce	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5190	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5357	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.6363	Ceres exchange	STATUS from Mars: F1 HELD for Cara	Tells Cara (no balance change)	Local access (1 s)	2.6365	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.9980	Ceres exchange	STATUS from Mars: F1 OPEN for Cara	Tells Cara (no balance change)	Local access (1 s)	2.9982	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
3.0674	Earth exchange	STATUS from Mars: F1 OPEN for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
62.5024	Bob	Schedule (learned locally at opening)	Publishes observation 1: index 110	Local access at Mars, sent h 62.5021	62.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
62.5024	Mars exchange	Signed observation from Bob	Records observation 1 = 110	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
122.5024	Bob	Schedule (learned locally at opening)	Publishes observation 2: index 120	Local access at Mars, sent h 122.5021	122.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
122.5024	Mars exchange	Signed observation from Bob	Records observation 2 = 120	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
182.5024	Bob	Schedule (learned locally at opening)	Publishes observation 3: index 130	Local access at Mars, sent h 182.5021	182.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
182.5024	Mars exchange	Signed observation from Bob	Records observation 3 = 130	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
242.5024	Bob	Schedule (learned locally at opening)	Publishes observation 4: index 135	Local access at Mars, sent h 242.5021	242.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
242.5024	Mars exchange	Signed observation from Bob	Records observation 4 = 135	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
302.5024	Bob	Schedule (learned locally at opening)	Publishes observation 5: index 125	Local access at Mars, sent h 302.5021	302.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
302.5024	Mars exchange	Signed observation from Bob	Records observation 5 = 125	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
326.5024	Mars exchange	Fixing cutoff; uses observation 5	Fixes at 125: long net \$50k; gross long \$130k, short \$30k	—	settles; claims backed	Alice \$70k at Earth, \$130k free at host; Cara \$20k at Ceres, \$30k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$130k; TRANSFER MA-X1 (payout) to Earth	New session (SYN h 326.5027); DATA Ma→A h 327.6171, A→Ea h 327.8825	328.1736	Alice \$70k at Earth, \$130k in transit; Cara \$20k at Ceres, \$30k free at host

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Cara \$30k; TRANSFER MA-X2 (payout) to Ceres	New session (SYN h 326.5029); DATA Ma→A h 327.5228, A→Ce h 327.7882	328.0321	Alice \$70k at Earth, \$130k in transit; Cara \$20k at Ceres, \$30k in transit
328.0321	Ceres exchange	Authentic new transfer MA-X2	Imports once; credits Cara \$30k (spendable now)	RECEIVED to Mars (backbone)	328.5422	Alice \$70k at Earth, \$130k in transit; Cara \$50k at Ceres
328.1736	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$130k (spendable now)	RECEIVED to Mars (backbone)	328.7309	Alice \$200k at Earth; Cara \$50k at Ceres
328.5422	Mars exchange	Ceres imported MA-X2	Marks transfer delivered; no balance change	—	—	Alice \$200k at Earth; Cara \$50k at Ceres
328.7309	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$200k at Earth; Cara \$50k at Ceres

Table A18: Complete trace of S1e: capped futures, falling path, $Q = 2,000$. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$80k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$80k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$80k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$70k at Earth, \$80k in transit; Cara \$100k at Ceres
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Cara \$80k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k in transit
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Cara \$80k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$70k at Earth, \$80k in transit; Cara \$20k at Ceres, \$80k free at host
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$80k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
1.9826	Ceres exchange	Mars imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
1.9829	Cara	Published geometry; own funding confirmed	Sends CINSTR $\times 2$ copies	Direct Ce $\rightarrow$ Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k free at host
2.1410	Mars exchange	Instruction + own funded balance	Holds Cara margin \$80k	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.1410	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma $\rightarrow$ Ce	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.1576	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR $\times 3$ copies	Direct Ea $\rightarrow$ Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$70k at Earth, \$80k free at host; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$80k	—	—	Alice \$70k at Earth, \$80k held; Cara \$20k at Ceres, \$80k held
2.5024	Mars exchange	Both margins held	Opens F1: escrow \$160k; maturity h 302.5024, fixing h 326.5024	STATUS (OPEN) records to Earth and Ceres (backbone)	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma $\rightarrow$ Ea	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma $\rightarrow$ Ce	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5190	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.5357	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.6363	Ceres exchange	STATUS from Mars: F1 HELD for Cara	Tells Cara (no balance change)	Local access (1 s)	2.6365	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
2.9980	Ceres exchange	STATUS from Mars: F1 OPEN for Cara	Tells Cara (no balance change)	Local access (1 s)	2.9982	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
3.0674	Earth exchange	STATUS from Mars: F1 OPEN for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
62.5024	Bob	Schedule (learned locally at opening)	Publishes observation 1: index 90	Local access at Mars, sent h 62.5021	62.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
62.5024	Mars exchange	Signed observation from Bob	Records observation 1 = 90	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
122.5024	Bob	Schedule (learned locally at opening)	Publishes observation 2: index 80	Local access at Mars, sent h 122.5021	122.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
122.5024	Mars exchange	Signed observation from Bob	Records observation 2 = 80	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
182.5024	Bob	Schedule (learned locally at opening)	Publishes observation 3: index 70	Local access at Mars, sent h 182.5021	182.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
182.5024	Mars exchange	Signed observation from Bob	Records observation 3 = 70	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
242.5024	Bob	Schedule (learned locally at opening)	Publishes observation 4: index 65	Local access at Mars, sent h 242.5021	242.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
242.5024	Mars exchange	Signed observation from Bob	Records observation 4 = 65	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
302.5024	Bob	Schedule (learned locally at opening)	Publishes observation 5: index 75	Local access at Mars, sent h 302.5021	302.5024	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
302.5024	Mars exchange	Signed observation from Bob	Records observation 5 = 75	Local access	—	Alice \$70k at Earth, \$80k margin; Cara \$20k at Ceres, \$80k margin
326.5024	Mars exchange	Fixing cutoff; uses observation 5	Fixes at 75: long net \$-50k; gross long \$30k, short \$130k	—	settles; claims backed	Alice \$70k at Earth, \$30k free at host; Cara \$20k at Ceres, \$130k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$30k; TRANSFER MA-X1 (payout) to Earth	New session (SYN h 326.5027); DATA Ma→A h 327.6171, A→Ea h 327.8825	328.1736	Alice \$70k at Earth, \$30k in transit; Cara \$20k at Ceres, \$130k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Cara \$130k; TRANSFER MA-X2 (payout) to Ceres	New session (SYN h 326.5029); DATA Ma→A h 327.5228, A→Ce h 327.7882	328.0321	Alice \$70k at Earth, \$30k in transit; Cara \$20k at Ceres, \$130k in transit
328.0321	Ceres exchange	Authentic new transfer MA-X2	Imports once; credits Cara \$130k (spendable now)	RECEIVED to Mars (backbone)	328.5422	Alice \$70k at Earth, \$30k in transit; Cara \$150k at Ceres
328.1736	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$30k (spendable now)	RECEIVED to Mars (backbone)	328.7309	Alice \$100k at Earth; Cara \$150k at Ceres
328.5422	Mars exchange	Ceres imported MA-X2	Marks transfer delivered; no balance change	—	—	Alice \$100k at Earth; Cara \$150k at Ceres
328.7309	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$100k at Earth; Cara \$150k at Ceres

Table A19: Complete trace of S1f: capped futures, boundary case $Q = 2,500$. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$100k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$50k at Earth, \$100k in transit; Cara \$100k at Ceres

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$100k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$50k at Earth, \$100k in transit; Cara \$100k at Ceres
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Cara \$100k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$50k at Earth, \$100k in transit; Cara \$100k in transit
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Cara \$100k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$50k at Earth, \$100k in transit; Cara \$100k free at host
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$50k at Earth, \$100k free at host; Cara \$100k free at host
1.9826	Ceres exchange	Mars imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k free at host
1.9829	Cara	Published geometry; own funding confirmed	Sends CINSTR ×2 copies	Direct Ce→Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$50k at Earth, \$100k free at host; Cara \$100k free at host
2.1410	Mars exchange	Instruction + own funded balance	Holds Cara margin \$100k	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.1410	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma→Ce	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.1576	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$50k at Earth, \$100k free at host; Cara \$100k held
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$100k	—	—	Alice \$50k at Earth, \$100k held; Cara \$100k held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5024	Mars exchange	Both margins held	Opens F1: escrow \$200k; maturity h 302.5024, fixing h 326.5024	STATUS (OPEN) records to Earth and Ceres (backbone)	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma→Ea	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Ceres	Backbone DATA Ma→Ce	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.5190	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.5357	Mars exchange	Same cinstr already decided (OPENED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.6363	Ceres exchange	STATUS from Mars: F1 HELD for Cara	Tells Cara (no balance change)	Local access (1 s)	2.6365	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
2.9980	Ceres exchange	STATUS from Mars: F1 OPEN for Cara	Tells Cara (no balance change)	Local access (1 s)	2.9982	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
3.0674	Earth exchange	STATUS from Mars: F1 OPEN for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
62.5024	Bob	Schedule (learned locally at opening)	Publishes observation 1: index 110	Local access at Mars, sent h 62.5021	62.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
62.5024	Mars exchange	Signed observation from Bob	Records observation 1 = 110	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
122.5024	Bob	Schedule (learned locally at opening)	Publishes observation 2: index 120	Local access at Mars, sent h 122.5021	122.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
122.5024	Mars exchange	Signed observation from Bob	Records observation 2 = 120	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
182.5024	Bob	Schedule (learned locally at opening)	Publishes observation 3: index 130	Local access at Mars, sent h 182.5021	182.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
182.5024	Mars exchange	Signed observation from Bob	Records observation 3 = 130	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
242.5024	Bob	Schedule (learned locally at opening)	Publishes observation 4: index 135	Local access at Mars, sent h 242.5021	242.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
242.5024	Mars exchange	Signed observation from Bob	Records observation 4 = 135	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
302.5024	Bob	Schedule (learned locally at opening)	Publishes observation 5: index 125	Local access at Mars, sent h 302.5021	302.5024	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
302.5024	Mars exchange	Signed observation from Bob	Records observation 5 = 125	Local access	—	Alice \$50k at Earth, \$100k margin; Cara \$100k margin
326.5024	Mars exchange	Fixing cutoff; uses observation 5	Fixes at 125: long net \$62.5k; gross long \$162.5k, short \$37.5k	—	settles; claims backed	Alice \$50k at Earth, \$162.5k free at host; Cara \$37.5k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$162.5k; TRANSFER MA-X1 (payout) to Earth	New session (SYN h 326.5027); DATA Ma→A h 327.6171, A→Ea h 327.8825	328.1736	Alice \$50k at Earth, \$162.5k in transit; Cara \$37.5k free at host
326.5024	Mars exchange	Instruction or fill; its own ledger	Debits Cara \$37.5k; TRANSFER MA-X2 (payout) to Ceres	New session (SYN h 326.5029); DATA Ma→A h 327.5228, A→Ce h 327.7882	328.0321	Alice \$50k at Earth, \$162.5k in transit; Cara \$37.5k in transit
328.0321	Ceres exchange	Authentic new transfer MA-X2	Imports once; credits Cara \$37.5k (spendable now)	RECEIVED to Mars (backbone)	328.5422	Alice \$50k at Earth, \$162.5k in transit; Cara \$37.5k at Ceres
328.1736	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$162.5k (spendable now)	RECEIVED to Mars (backbone)	328.7309	Alice \$212.5k at Earth; Cara \$37.5k at Ceres
328.5422	Mars exchange	Ceres imported MA-X2	Marks transfer delivered; no balance change	—	—	Alice \$212.5k at Earth; Cara \$37.5k at Ceres
328.7309	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$212.5k at Earth; Cara \$37.5k at Ceres

Table A20: Complete trace of S1g: capped futures, $Q = 2,501$ (Cara cannot fund; Alice is refunded). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$100.04k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$100.04k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$100.04k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
0.0003	Ceres exchange	insufficient available ND: have 100000, need 100040	Rejects Cara FUND F1-Cara-fund	—	—	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$100.04k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
2.2625	Alice	Published geometry; own funding confirmed	Sends CINSTR ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
2.5024	Mars exchange	Instruction + own funded balance	Holds Alice margin \$100.04k	—	—	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
2.5024	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma→Ea	—	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
2.5190	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
2.5357	Mars exchange	Same cinstr already decided (HELD)	Ignores the copy; returns the original result	—	—	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres
3.0674	Earth exchange	STATUS from Mars: F1 HELD for Alice	Tells Alice (no balance change)	Local access (1 s)	3.0677	Alice \$49.96k at Earth, \$100.04k held; Cara \$100k at Ceres

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
168.0000	Mars exchange	opening deadline passed	Releases Alice margin \$100.04k	—	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
168.0000	Mars exchange	both instructions not held by deadline	Rejects F1	—	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
168.0000	Mars exchange	Contract state changed for remote clients	Queues STATUS record (1 entry) for Earth	Backbone DATA Ma→Ea	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
168.5604	Earth exchange	STATUS from Mars: F1 REJECTED (deadline) for Alice	Tells Alice (no balance change)	Local access (1 s)	168.5607	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
169.0000	Alice	no OPEN report by the opening deadline	withdraw after deadline	—	—	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
169.0000	Alice	Published geometry; own funding confirmed	Sends WITHDRAW ×3 copies	Direct Ea→Ma, launches 169.0000, 169.0167, 169.0333; p(loss) 0.126 each, all lost 2.0e-03	169.2336 (first)	Alice \$49.96k at Earth, \$100.04k free at host; Cara \$100k at Ceres
169.2336	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$100.04k; TRANSFER MA-X1 (withdrawal) to Earth	Session from h 0.0006; DATA Ma→A h 169.2338, A→Ea h 169.4928	169.7939	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
169.2502	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
169.2669	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$49.96k at Earth, \$100.04k in transit; Cara \$100k at Ceres
169.7939	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$100.04k (spendable now)	RECEIVED to Mars (backbone)	170.3549	Alice \$150k at Earth; Cara \$100k at Ceres
170.3549	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$150k at Earth; Cara \$100k at Ceres

Table A21: Complete trace of S1h: capped futures, $Q = 4,000$ (both fundings refused at home). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Own Earth balance	Funds own Mars account: \$160k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Earth exchange	insufficient available ND: have 150000, need 160000	Rejects Alice FUND F1-Alice-fund	—	—	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Cara	Own Ceres balance	Funds own Mars account: \$160k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$150k at Earth; Cara \$100k at Ceres
0.0003	Ceres exchange	insufficient available ND: have 100000, need 160000	Rejects Cara FUND F1-Cara-fund	—	—	Alice \$150k at Earth; Cara \$100k at Ceres

Table A22: Complete trace of S1i: repeat trading on a \$10,000 Mars account. Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Earth exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ea→Ma, batched with queued records	1.6966	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Alice	Own Earth balance	Funds own Mars account: \$10k	Local access at Earth, sent h 0.0000	0.0003	Alice \$150k at Earth; Bob \$50k at Mars, 3,000 Ares home
0.0003	Earth exchange	Instruction or fill; its own ledger	Debits Alice \$10k; TRANSFER EA-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ea→A h 1.1317, A→Ma h 1.4448	1.6966	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 3,000 Ares home
0.0003	Bob	Own inventory at the market	Sell 10 Ares limit \$97 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob RT-s1; holds 10 Ares	—	—	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 2,990 Ares home, 10 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 5 Ares limit \$100 GTC	Local access at Mars, sent h 0.0001	0.0003	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 2,990 Ares home, 10 Ares held
0.0003	Mars exchange	Its own available balances	Admits Bob RT-s2; holds 5 Ares	—	—	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
1.6966	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 2)	Backbone Ma→Ea, batched with queued records	2.2622	Alice \$140k at Earth, \$10k in transit; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
1.6966	Mars exchange	Authentic new transfer EA-X1	Imports once; credits Alice \$10k (spendable now)	RECEIVED to Earth (backbone)	2.2622	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.2622	Earth exchange	MDS from Mars	Posts Ares snapshot to its board (0.57 h old); answers waiting QUOTEs	Local access (1 s)	2.2625	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.2622	Earth exchange	Mars imported EA-X1	Marks transfer delivered; no balance change	—	—	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.2625	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$97 × 10, 0.57 h old	—	—	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.2625	Alice	Published geometry; own funding confirmed	Sends ORDER o1+o2 ×3 copies	Direct Ea→Ma, launches 2.2625, 2.2791, 2.2958; p(loss) 0.129 each, all lost 2.1e-03	2.5024 (first)	Alice \$140k at Earth, \$10k free at host; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.5024	Mars exchange	Its own available balances	Admits Alice RT-o1; holds \$4k	—	—	Alice \$140k at Earth, \$6k free at host, \$4k held; Bob \$50k at Mars, 2,985 Ares home, 15 Ares held
2.5024	Mars exchange	Both holds in its own ledger	Fill 10 Ares @ \$97: Alice buys from Bob; improvement \$30	—	local, atomic	Alice \$140k at Earth, \$6.03k free at host, \$3k held, 10 Ares at market; Bob \$50.97k at Mars, 2,985 Ares home, 5 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.5024	Mars exchange	Both holds in its own ledger	Fill 5 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5024	Mars exchange	Its own available balances	Admits Alice RT-o2; holds \$4k	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5190	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5190	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5357	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
2.5357	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
8.2625	Alice	Published geometry; own funding confirmed	Sends CANCEL o2 ×3 copies	Direct Ea→Ma, launches 8.2625, 8.2791, 8.2958; p(loss) 0.129 each, all lost 2.1e-03	8.5021 (first)	Alice \$140k at Earth, \$2.03k free at host, \$6.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
8.5021	Mars exchange	Order state	Cancelled RT-o2; releases \$4k	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
8.5188	Mars exchange	RT-o2 already CANCELLED	Cancel has no effect	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
8.5355	Mars exchange	RT-o2 already CANCELLED	Cancel has no effect	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
14.2625	Alice	Published geometry; own funding confirmed	Sends ORDER o3 ×3 copies	Direct Ea→Ma, launches 14.2625, 14.2791, 14.2958; p(loss) 0.129 each, all lost 2.1e-03	14.5019 (first)	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 15 Ares at market; Bob \$51.47k at Mars, 2,985 Ares home
14.5019	Mars exchange	Its own available balances	Admits Alice RT-o3; holds 10 Ares	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$51.47k at Mars, 2,985 Ares home
14.5186	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$51.47k at Mars, 2,985 Ares home
14.5353	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$51.47k at Mars, 2,985 Ares home
15.2627	Bob	Own inventory at the market	Buy 10 Ares limit \$101 IOC	Local access at Mars, sent h 15.2625	15.2627	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$51.47k at Mars, 2,985 Ares home
15.2627	Mars exchange	Its own available balances	Admits Bob RT-b1; holds \$1.01k	—	—	Alice \$140k at Earth, \$6.03k free at host, \$2.5k held, 5 Ares at market, 10 Ares held; Bob \$50.46k at Mars, \$1.01k held, 2,985 Ares home
15.2627	Mars exchange	Both holds in its own ledger	Fill 10 Ares @ \$101: Bob buys from Alice	—	local, atomic	Alice \$140k at Earth, \$7.04k free at host, \$2.5k held, 5 Ares at market; Bob \$50.46k at Mars, 2,995 Ares home
26.2625	Alice	Published geometry; own funding confirmed	Sends CANCEL o1 ×3 copies	Direct Ea→Ma, launches 26.2625, 26.2791, 26.2958; p(loss) 0.128 each, all lost 2.1e-03	26.5015 (first)	Alice \$140k at Earth, \$7.04k free at host, \$2.5k held, 5 Ares at market; Bob \$50.46k at Mars, 2,995 Ares home
26.3627	Bob	Own inventory at the market	Sell 5 Ares limit \$99 IOC	Local access at Mars, sent h 26.3625	26.3627	Alice \$140k at Earth, \$7.04k free at host, \$2.5k held, 5 Ares at market; Bob \$50.46k at Mars, 2,995 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
26.3627	Mars exchange	Its own available balances	Admits Bob RT-s3; holds 5 Ares	—	—	Alice \$140k at Earth, \$7.04k free at host, \$2.5k held, 5 Ares at market; Bob \$50.46k at Mars, 2,990 Ares home, 5 Ares held
26.3627	Mars exchange	Both holds in its own ledger	Fill 5 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$140k at Earth, \$7.04k free at host, \$2k held, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
26.5015	Mars exchange	Order state	Cancelled RT-o1; releases \$2k	—	—	Alice \$140k at Earth, \$9.04k free at host, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
26.5181	Mars exchange	RT-o1 already CANCELLED	Cancel has no effect	—	—	Alice \$140k at Earth, \$9.04k free at host, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
26.5348	Mars exchange	RT-o1 already CANCELLED	Cancel has no effect	—	—	Alice \$140k at Earth, \$9.04k free at host, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
50.2625	Alice	Published geometry; own funding confirmed	Sends WITHDRAW cash+ARES $\times$ 3 copies	Direct Ea $\rightarrow$ Ma, launches 50.2625, 50.2791, 50.2958; p(loss) 0.128 each, all lost 2.1e-03	50.5006 (first)	Alice \$140k at Earth, \$9.04k free at host, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
50.5006	Mars exchange	Instruction or fill; its own ledger	Debits Alice \$9.04k; TRANSFER MA-X1 (withdrawal) to Earth	Session from h 0.0006; DATA Ma $\rightarrow$ A h 50.5009, A $\rightarrow$ Ea h 50.7551	51.0642	Alice \$140k at Earth, \$9.04k in transit, 10 Ares at market; Bob \$50.96k at Mars, 2,990 Ares home
50.5006	Mars exchange	Instruction or fill; its own ledger	Debits Alice 10 Ares; TRANSFER MA-X2 (withdrawal) to Earth	Session from h 0.0006; DATA Ma $\rightarrow$ A h 50.5009, A $\rightarrow$ Ea h 50.7551	51.0642	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
50.5172	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
50.5172	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
50.5339	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
50.5339	Mars exchange	Same withdraw already decided	Ignores the copy; returns the original result	—	—	Alice \$140k at Earth, \$9.04k in transit, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
51.0642	Earth exchange	Authentic new transfer MA-X1	Imports once; credits Alice \$9.04k (spendable now)	RECEIVED to Mars (backbone)	51.6284	Alice \$149.04k at Earth, 10 Ares in transit; Bob \$50.96k at Mars, 2,990 Ares home
51.0642	Earth exchange	Authentic new transfer MA-X2	Imports once; credits Alice 10 Ares (spendable now)	RECEIVED to Mars (backbone)	51.6284	Alice \$149.04k at Earth, 10 Ares home; Bob \$50.96k at Mars, 2,990 Ares home
51.6284	Mars exchange	Earth imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$149.04k at Earth, 10 Ares home; Bob \$50.96k at Mars, 2,990 Ares home
51.6284	Mars exchange	Earth imported MA-X2	Marks transfer delivered; no balance change	—	—	Alice \$149.04k at Earth, 10 Ares home; Bob \$50.96k at Mars, 2,990 Ares home

E2. Probability and traffic

Direct service (exact). Each copy is lost independently with probability $p = 1 - e^{-0.08d}$, where d is the distance in AU at that copy’s own launch. Copies are sent 60 s apart, and we check that the path is open for every copy, so they are never blocked by the same closure. Incidents are excluded from these figures. The probability that all copies are lost is the product of the p_i . A lost order is not a financial failure: the cash stays available at the host until a fallback copy arrives. Table A23 lists every direct message in S1a to S1e; Table A26 in Section E2+ extends it to all nine runs. The traces use the first copy’s arrival. Each later copy adds 60 s and raises the probability, so Alice’s order in S1a has arrived with probability 0.871 at the traced time and 0.9979 2 minutes later. If no copy arrives, the client resends with the same identifier (R6). HELD and OPEN status reports are not direct messages: they travel to the client’s home exchange as STATUS records over the backbone, and are counted with the backbone packets.

Backbone. Table A24 covers every hop of every record in S1a to S1e; Table A29 in Section E2+ lists every launch of every S1 run individually. For each hop a launch is unconfirmed with probability $1 - (1 - p_{\text{data}})(1 - p_{\text{receipt}})$. P(no retry) is the probability that every hop of the record is confirmed at its first launch, which is the case shown in the traces. A hop is abandoned after four unconfirmed launches. Launch k is unconfirmed with probability $u_k = 1 - e^{-0.02d_k} e^{-0.02r_k}$, where d_k and r_k are the path lengths of its data and of the hop receipt, so the hop is abandoned with probability $u_1 u_2 u_3 u_4$. P(ab.) is the chance that this happens on some hop, with each retry launched when the hop timer sets it ($t_{k+1} = t_k +$ twice the flight time +1 h) and each u_k computed from its own geometry (hop_bound.py). Queues or quota may delay a retry. If each retry may be delayed by up to 24 hours, launch k lies in a window of $(k - 1) \cdot 24$ h, no path length grows faster than

Table A23: Direct-service messages in the S1 runs (exact). “First h is the arrival time of the first copy, which is the time used in the traces; “P by then is the probability that the message has arrived by that time. P(arrives) is the probability that at least one copy arrives, by the last copy’s arrival.

Run	Sender	Path	Message	Copies	p each	First h	P by then	Last h	P(arrives)
S1a	Alice	Ea→Ma	ORDER	3	0.1289	2.502	0.8711	2.536	0.99786
S1c	Eve	Ur→Ea	ORDER	8	0.7815	13.188	0.2185	13.304	0.86082
S1d	Cara	Ce→Ma	CINSTR	2	0.0867	2.141	0.9133	2.158	0.99248
S1d	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.99786
S1i	Alice	Ea→Ma	ORDER o1 + o2	3	0.1289	2.502	0.8711	2.536	0.99786
S1i	Alice	Ea→Ma	CANCEL o2	3	0.1288	8.502	0.8712	8.535	0.99786
S1i	Alice	Ea→Ma	ORDER o3	3	0.1287	14.502	0.8713	14.535	0.99787
S1i	Alice	Ea→Ma	CANCEL o1	3	0.1284	26.501	0.8716	26.535	0.99788
S1i	Alice	Ea→Ma	WITHDRAW cash + ARES	3	0.1280	50.501	0.8720	50.534	0.99790

0.0029 AU/h (twice Mercury’s top speed, with margin), and u grows with both lengths, so evaluating u_k at $d_k + 0.0029 \cdot (k - 1) \cdot 24$ (and the same for r_k) gives the column Bound. Every window is also certified open: the clearance at its start exceeds 0.10 AU + 0.00146 AU/h times its length (the clearance Lipschitz bound of E4). *The bound is conditional on that 24-hour limit.* It does not cover a retry held longer, for example by a closure of the hop that opens after the first launch or by congestion that fills the link for more than a day; for such cases we give no per-hop bound, and the Monte Carlo results below are the evidence. Holding the first launch’s geometry for all four launches, our earlier approximation, is within 0.21% of the exact value on every hop. P(never) is the chance that the data never crosses some hop (p_{data}^4 per hop). An abandoned hop leads to endpoint retries and resubmission; the money stays in transit, owned by the client.

Table A24: Backbone loss for every record of S1a to S1e, over all of its hops. P(ab.), the chance that some hop is abandoned, is exact for retries at the hop timer; Bound holds only if no retry is delayed by more than 24 hours. S1d/e: S1e uses the same records and times as S1d. Ea = Earth, Ma = Mars, Ce = Ceres, Ur = Uranus.

Run	Record	Hops	Max d	Launch	Max p	unc.	P(no retry)	P(ab.)	Bound	P(never)
S1a	Funding Ea→Ma	4	2.26	1.13		0.0863	0.7221	1.6e-04	1.9e-04	1.1e-05
S1a	Delivery Ma→Ea	4	2.25	2.50		0.0862	0.7221	1.6e-04	1.9e-04	1.1e-05
S1b	Funding Ea→Ce	4	2.26	1.11		0.0863	0.7258	1.5e-04	1.8e-04	1.0e-05
S1b	Delivery Ce→Ea	4	2.25	2.61		0.0862	0.7259	1.5e-04	1.8e-04	1.0e-05
S1c	Delivery Ea→Ur	4	16.77	13.19		0.4886	0.2185	1.1e-01	1.1e-01	1.3e-02
S1c	Funding Ur→Ea	4	16.77	5.28		0.4886	0.2184	1.1e-01	1.1e-01	1.3e-02
S1d/e	Funding Ea→Ma	4	2.26	1.13		0.0863	0.7221	1.6e-04	1.9e-04	1.1e-05
S1d/e	Payout Ma→Ea	4	2.10	327.62		0.0806	0.7255	1.4e-04	1.7e-04	9.7e-06
S1d/e	Payout Ma→Ce	4	1.91	327.52		0.0736	0.7456	1.0e-04	1.2e-04	6.8e-06
S1d/e	Funding Ce→Ma	4	1.82	0.99		0.0701	0.7517	9.0e-05	1.1e-04	6.0e-06

Monte Carlo under random loss. Each run uses an independent seed, draws every launch from the brief’s loss law and follows our client policy. Table A25 ties each traced time to its probability: the no-loss time is a best case, met only when every launch succeeds first time. Figure A1 shows the full distributions.

Futures opening and link capacity. The contract opened in 100.00% of the lossy runs. The median opening time was 3.63 h and the 99th percentile 9.94 h, well before the 168-hour deadline. The longest queue on any directed link in any S1 run was 4 packets, against a limit of 10,000, and serialization never delayed a packet by more than a few seconds.

Table A25: Monte Carlo results under random loss (2,000 runs each; every run completed). Times in hours. $P(\leq \text{tr.})$ and $P(+1\text{h})$ are the probabilities of completing by the no-loss trace time and within one hour of it. Quota columns give the highest rolling 24-hour use in any run: backbone originations by one exchange (limit 66) and direct packets by one client (limit 12). The last two columns count the runs that needed an application resubmission or a client fallback.

Run	Trace h	$P(\leq \text{tr.})$	$P(+1\text{h})$	Median	p90	p99	Max	Pkts	BB q.	Dir. q.	Res.	Fb.
S1a	3.07	0.53	0.60	3.07	4.71	7.70	28.21	49.8	3	6	0	1
S1b	3.16	0.50	0.62	3.16	4.81	7.76	14.04	49.8	3	6	0	9
S1c	15.82	0.02	0.09	28.76	50.74	71.48	104.58	69.2	3	12	0	414
S1d	328.17	0.21	0.24	329.70	331.42	335.62	340.26	149.1	5	6	0	21
S1e	328.17	0.21	0.24	329.70	331.42	335.62	340.26	149.1	5	6	0	21

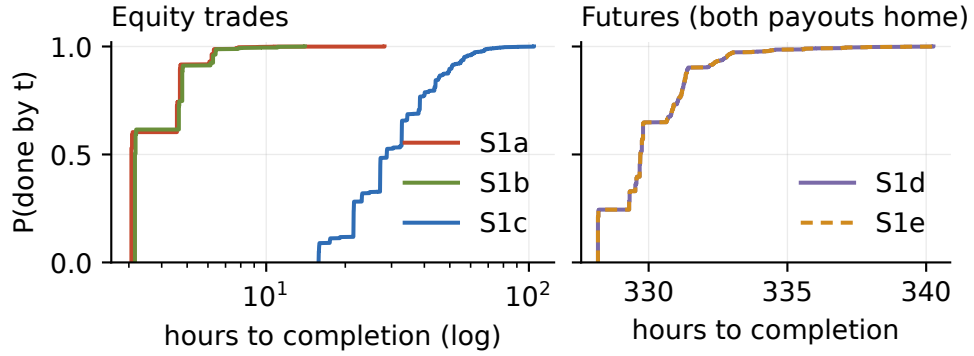


Figure A1: Completion-time distributions under random loss (empirical CDF of the Monte Carlo runs in Table A25). Left: the three equity trades, log scale. Right: the futures, both payouts home.

E2+. Every message and every launch

Table A26 lists every direct message of every S1 run, including the S1b order to Ceres, the S1g withdrawal after the deadline and the S1i batches. Table A29 lists every backbone launch except the hop receipts, which are the second factor of each row’s probability. Table A27 summarises them per run.

Table A26: Every direct message in every S1 run (exact). “First h” is the arrival of the first copy, the time used in the traces, and “P by then” the probability that the message has arrived by then. P(arrives) is the probability that at least one copy arrives, reached at the last copy’s arrival. S1h sends no direct message: both fundings are refused at home.

Run	Sender	Path	Message	Copies	p each	First h	P by then	Last h	P(arrives)
S1a	Alice	Ea→Ma	ORDER	3	0.1289	2.502	0.8711	2.536	0.997859
S1b	Alice	Ea→Ce	ORDER	3	0.1974	2.608	0.8026	2.642	0.992306
S1c	Eve	Ur→Ea	ORDER	8	0.7815	13.188	0.2185	13.304	0.860817
S1d	Cara	Ce→Ma	CINSTR	2	0.0867	2.141	0.9133	2.158	0.992476
S1d	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.997859
S1e	Cara	Ce→Ma	CINSTR	2	0.0867	2.141	0.9133	2.158	0.992476
S1e	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.997859
S1f	Cara	Ce→Ma	CINSTR	2	0.0867	2.141	0.9133	2.158	0.992476
S1f	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.997859
S1g	Alice	Ea→Ma	CINSTR	3	0.1289	2.502	0.8711	2.536	0.997859
S1g	Alice	Ea→Ma	WITHDRAW	3	0.1257	169.234	0.8743	169.267	0.998015
S1i	Alice	Ea→Ma	ORDER o1 + o2	3	0.1289	2.502	0.8711	2.536	0.997859
S1i	Alice	Ea→Ma	CANCEL o2	3	0.1288	8.502	0.8712	8.535	0.997865
S1i	Alice	Ea→Ma	ORDER o3	3	0.1287	14.502	0.8713	14.535	0.997870
S1i	Alice	Ea→Ma	CANCEL o1	3	0.1284	26.501	0.8716	26.535	0.997881
S1i	Alice	Ea→Ma	WITHDRAW cash + ARES	3	0.1280	50.501	0.8720	50.534	0.997904

Table A27: Per-run summary of Tables A26 and A29. The first probability is the chance that every backbone hop is confirmed at its first launch (direct copies not included); the second is the chance that at least one hop is abandoned and endpoint recovery takes over (exact, then the 24-hour-delay bound).

Run	Direct msgs	Backbone launches	P(no backbone retry)	P(some hop abandoned)	Bound
S1a	1	22	0.1668	8.7e-04	1.0e-03
S1b	1	22	0.1716	8.4e-04	1.0e-03
S1c	1	22	0.0002	4.8e-01	4.8e-01
S1d	2	68	0.0056	2.1e-03	2.5e-03
S1e	2	68	0.0056	2.1e-03	2.5e-03
S1f	2	68	0.0056	2.1e-03	2.5e-03
S1g	2	30	0.0877	1.2e-03	1.4e-03
S1h	0	0	1.0000	0.0e+00	0.0e+00
S1i	5	22	0.1671	8.7e-04	1.0e-03

What the Monte Carlo shows and does not show. Every lossy run reached its end point within the run (Table A28). With 2,000 runs and no failure, the probability of failing within 1,500 hours is below 0.15% at 95% confidence; the simulations alone cannot show more. Completion with probability one rests on the rules, not on the sample. A transfer is resubmitted after every delivery-unknown, without limit. Each attempt crosses a route of at most three links whose lengths are bounded for all time (E4), so it succeeds with probability at least some $q > 0$, and after k attempts it is still undelivered with probability at most $(1 - q)^k$, which goes to zero. A client request is resent by R6 until it is 30 days old. A completion *time* is never certain: every table gives a distribution, and the no-loss trace time is its best case.

Table A28: Monte Carlo under random loss for every S1 run with network traffic (2,000 runs each). The last column is the 95% upper confidence bound on the probability of not reaching the end point within the run (1,500 hours) when no run failed (rule of three, $3/n$). S1h has no network traffic.

Run	End point	Done	Median h	p90 h	p99 h	Max h	95\% bound
S1a	completion	2000/2000	3.07	4.71	7.70	28.21	0.15%
S1b	completion	2000/2000	3.16	4.81	7.76	14.04	0.15%
S1c	completion	2000/2000	28.76	50.74	71.48	104.58	0.15%
S1d	completion	2000/2000	329.70	331.42	335.62	340.26	0.15%
S1e	completion	2000/2000	329.70	331.42	335.62	340.26	0.15%
S1f	completion	2000/2000	329.70	331.42	335.62	340.26	0.15%
S1g	margin home	2000/2000	169.79	170.16	171.77	177.40	0.15%
S1i	completion	2000/2000	51.06	52.71	55.70	77.70	0.15%

Table A29: Every backbone launch of every S1 run except hop receipts, with its per-launch probabilities. “p unconf.” is the chance that the launch is not confirmed: the data or its hop receipt is lost, $1 - (1 - p_{\text{data}})(1 - p_{\text{receipt}})$. P(aband.) is the chance that the hop is abandoned after four unconfirmed launches, $u_1 u_2 u_3 u_4$ with each retry at the hop timer and its own geometry; Bound holds only if no retry is delayed by more than 24 hours (Section E2). TR = TRANSFER, RCV = RECEIVED.

Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1a	0.001	Ea→A	SYN	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1a	0.314	A→Ma	SYN	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	0.566	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	0.818	A→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1a	1.131	Ea→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1a	1.132	Ea→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1a	1.445	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	1.445	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	1.697	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	1.697	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	1.949	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	1.950	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	2.263	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	2.503	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	2.576	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	2.755	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	3.068	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	3.068	Ea→A	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1a	3.381	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	3.381	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	3.634	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1a	3.886	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	0.001	Ea→A	SYN	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1b	0.314	A→Ce	SYN	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	0.557	Ce→A	SYNACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	0.800	A→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1b	1.113	Ea→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1b	1.114	Ea→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1b	1.427	A→Ce	ACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	1.427	A→Ce	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	1.670	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	1.670	Ce→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	1.913	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	1.914	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05

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Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1b	2.227	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	2.540	A→Ce	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	2.609	Ce→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	2.852	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	3.165	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	3.165	Ea→A	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1b	3.478	A→Ce	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	3.478	A→Ce	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	3.722	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1b	3.965	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1c	0.001	Ur→A	SYN	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	2.325	A→Ea	SYN	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1c	2.638	Ea→A	SYNACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1c	2.951	A→Ur	SYNACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	5.276	Ur→A	ACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	5.277	Ur→A	DATA (RCV+TR)	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	7.601	A→Ea	ACK	2.25	0.0440	0.0440	0.0861	5.5e-05	6.5e-05
S1c	7.601	A→Ea	DATA (RCV+TR)	2.25	0.0440	0.0440	0.0861	5.5e-05	6.5e-05
S1c	7.914	Ea→A	DACK	2.25	0.0440	0.0440	0.0861	5.5e-05	6.5e-05
S1c	7.914	Ea→A	DATA (RCV+TR)	2.25	0.0440	0.0440	0.0861	5.5e-05	6.5e-05
S1c	8.227	A→Ur	DACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	8.227	A→Ur	DATA (RCV+TR)	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	10.552	Ur→A	DACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	12.876	A→Ea	DACK	2.25	0.0440	0.0440	0.0860	5.5e-05	6.5e-05
S1c	13.188	Ea→A	DATA (RCV+TR)	2.25	0.0440	0.0440	0.0860	5.5e-05	6.5e-05
S1c	13.500	A→Ur	DATA (RCV+TR)	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	15.825	Ur→A	DACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	15.825	Ur→A	DATA (RCV+TR)	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1c	18.149	A→Ea	DACK	2.25	0.0439	0.0439	0.0859	5.4e-05	6.5e-05
S1c	18.150	A→Ea	DATA (RCV+TR)	2.25	0.0439	0.0439	0.0859	5.4e-05	6.5e-05
S1c	18.462	Ea→A	DACK	2.25	0.0439	0.0439	0.0859	5.4e-05	6.5e-05
S1c	18.774	A→Ur	DACK	16.77	0.2849	0.2849	0.4886	5.7e-02	5.8e-02
S1d	0.001	Ea→A	SYN	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1d	0.001	Ce→A	SYN	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	0.244	A→Ma	SYN	1.82	0.0357	0.0357	0.0700	2.4e-05	3.0e-05
S1d	0.314	A→Ma	SYN	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	0.496	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	0.566	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	0.748	A→Ce	SYNACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	0.818	A→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1d	0.992	Ce→A	ACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	0.992	Ce→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	1.131	Ea→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1d	1.132	Ea→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1d	1.235	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.235	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.445	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.445	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.487	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.488	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.697	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	1.697	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05

(continued)

Run	Launch	h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1d	1.740	A→Ce	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	1.740	A→Ce	DATA (RCV+TR)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	1.949	A→Ea	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	1.950	A→Ea	DATA (RCV+TR)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	1.983	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	2.141	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.226	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.263	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	2.394	A→Ce	DATA (STATUS)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	2.503	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.503	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.576	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.637	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	2.755	A→Ea	DATA (STATUS)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	2.755	A→Ce	DATA (STATUS)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	2.880	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	2.999	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1d	3.068	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1d	3.242	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	3.381	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1d	326.503	Ma→A	SYN		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	326.503	Ma→A	SYN		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	326.768	A→Ea	SYN		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	326.768	A→Ce	SYN		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	327.013	Ce→A	SYNACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	327.060	Ea→A	SYNACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	327.257	A→Ma	SYNACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.351	A→Ma	SYNACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.523	Ma→A	ACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.523	Ma→A	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.617	Ma→A	ACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.617	Ma→A	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	327.788	A→Ce	ACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	327.788	A→Ce	DATA (RCV+TR)		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	327.882	A→Ea	ACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	327.882	A→Ea	DATA (RCV+TR)		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	328.033	Ce→A	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	328.033	Ce→A	DATA (RCV+TR)		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	328.174	Ea→A	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	328.174	Ea→A	DATA (RCV+TR)		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1d	328.277	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.277	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.466	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.466	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.543	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.731	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1d	328.808	A→Ce	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1d	328.997	A→Ea	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.0e-05
S1e	0.001	Ea→A	SYN		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1e	0.001	Ce→A	SYN		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	0.244	A→Ma	SYN		1.82	0.0357	0.0357	0.0700	2.4e-05	3.0e-05
S1e	0.314	A→Ma	SYN		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05

(continued)

Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1e	0.496	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	0.566	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	0.748	A→Ce	SYNACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	0.818	A→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1e	0.992	Ce→A	ACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	0.992	Ce→A	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	1.131	Ea→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1e	1.132	Ea→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1e	1.235	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.235	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.445	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.445	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.487	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.488	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.697	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.697	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	1.740	A→Ce	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	1.740	A→Ce	DATA (RCV+TR)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	1.949	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	1.950	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	1.983	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	2.141	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.226	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.263	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	2.394	A→Ce	DATA (STATUS)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	2.503	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.503	Ma→A	DATA (STATUS)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.576	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.637	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	2.755	A→Ea	DATA (STATUS)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	2.755	A→Ce	DATA (STATUS)	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	2.880	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	2.999	Ce→A	DACK	1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1e	3.068	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1e	3.242	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	3.381	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1e	326.503	Ma→A	SYN	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	326.503	Ma→A	SYN	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	326.768	A→Ea	SYN	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	326.768	A→Ce	SYN	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	327.013	Ce→A	SYNACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	327.060	Ea→A	SYNACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	327.257	A→Ma	SYNACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.351	A→Ma	SYNACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.523	Ma→A	ACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.523	Ma→A	DATA (RCV+TR)	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.617	Ma→A	ACK	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.617	Ma→A	DATA (RCV+TR)	1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	327.788	A→Ce	ACK	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	327.788	A→Ce	DATA (RCV+TR)	1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	327.882	A→Ea	ACK	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	327.882	A→Ea	DATA (RCV+TR)	2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05

(continued)

Run	Launch	h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1e	328.033	Ce→A	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	328.033	Ce→A	DATA (RCV+TR)		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	328.174	Ea→A	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	328.174	Ea→A	DATA (RCV+TR)		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1e	328.277	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.277	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.466	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.466	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.543	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.731	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1e	328.808	A→Ce	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1e	328.997	A→Ea	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.0e-05
S1f	0.001	Ea→A	SYN		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1f	0.001	Ce→A	SYN		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	0.244	A→Ma	SYN		1.82	0.0357	0.0357	0.0700	2.4e-05	3.0e-05
S1f	0.314	A→Ma	SYN		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	0.496	Ma→A	SYNACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	0.566	Ma→A	SYNACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	0.748	A→Ce	SYNACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	0.818	A→Ea	SYNACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1f	0.992	Ce→A	ACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	0.992	Ce→A	DATA (RCV+TR)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	1.131	Ea→A	ACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1f	1.132	Ea→A	DATA (RCV+TR)		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1f	1.235	A→Ma	ACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.235	A→Ma	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.445	A→Ma	ACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.445	A→Ma	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.487	Ma→A	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.488	Ma→A	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.697	Ma→A	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.697	Ma→A	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	1.740	A→Ce	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	1.740	A→Ce	DATA (RCV+TR)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	1.949	A→Ea	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	1.950	A→Ea	DATA (RCV+TR)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	1.983	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	2.141	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.226	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.263	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	2.394	A→Ce	DATA (STATUS)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	2.503	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.503	Ma→A	DATA (STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.576	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.637	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	2.755	A→Ea	DATA (STATUS)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	2.755	A→Ce	DATA (STATUS)		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	2.880	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	2.999	Ce→A	DACK		1.75	0.0344	0.0344	0.0676	2.1e-05	2.6e-05
S1f	3.068	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1f	3.242	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1f	3.381	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05

(continued)

Run	Launch	h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1f	326.503	Ma→A	SYN		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	326.503	Ma→A	SYN		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	326.768	A→Ea	SYN		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	326.768	A→Ce	SYN		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	327.013	Ce→A	SYNACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	327.060	Ea→A	SYNACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	327.257	A→Ma	SYNACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.351	A→Ma	SYNACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.523	Ma→A	ACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.523	Ma→A	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.617	Ma→A	ACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.617	Ma→A	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	327.788	A→Ce	ACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	327.788	A→Ce	DATA (RCV+TR)		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	327.882	A→Ea	ACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	327.882	A→Ea	DATA (RCV+TR)		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	328.033	Ce→A	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	328.033	Ce→A	DATA (RCV+TR)		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	328.174	Ea→A	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	328.174	Ea→A	DATA (RCV+TR)		2.10	0.0411	0.0411	0.0806	4.2e-05	5.1e-05
S1f	328.277	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.277	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.466	A→Ma	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.466	A→Ma	DATA (RCV+TR)		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.543	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.731	Ma→A	DACK		1.91	0.0375	0.0375	0.0736	2.9e-05	3.6e-05
S1f	328.808	A→Ce	DACK		1.76	0.0346	0.0346	0.0680	2.1e-05	2.7e-05
S1f	328.997	A→Ea	DACK		2.10	0.0411	0.0411	0.0806	4.2e-05	5.0e-05
S1g	0.001	Ea→A	SYN		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1g	0.314	A→Ma	SYN		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	0.566	Ma→A	SYNACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	0.818	A→Ea	SYNACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1g	1.131	Ea→A	ACK		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1g	1.132	Ea→A	DATA (RCV+TR)		2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1g	1.445	A→Ma	ACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	1.445	A→Ma	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	1.697	Ma→A	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	1.697	Ma→A	DATA (RCV+TR)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	1.949	A→Ea	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	1.950	A→Ea	DATA (RCV+TR)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	2.263	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	2.503	Ma→A	DATA (RCV+STATUS)		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	2.576	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	2.755	A→Ea	DATA (RCV+STATUS)		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	3.068	Ea→A	DACK		2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1g	3.381	A→Ma	DACK		1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1g	168.000	Ma→A	DATA (STATUS)		1.86	0.0366	0.0366	0.0718	2.7e-05	3.3e-05
S1g	168.259	A→Ea	DATA (STATUS)		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	168.561	Ea→A	DACK		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	168.863	A→Ma	DACK		1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	169.234	Ma→A	DATA (TR)		1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	169.493	A→Ea	DATA (TR)		2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05

(continued)

Run	Launch h	Hop	Packet	d (AU)	p data	p rcpt	p unconf.	P(aband.)	Bound
S1g	169.794	Ea→A	DACK	2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	169.795	Ea→A	DATA (RCV+STATUS)	2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1g	170.096	A→Ma	DACK	1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	170.096	A→Ma	DATA (RCV+STATUS)	1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	170.355	Ma→A	DACK	1.86	0.0366	0.0366	0.0719	2.7e-05	3.3e-05
S1g	170.614	A→Ea	DACK	2.17	0.0425	0.0425	0.0832	4.8e-05	5.7e-05
S1i	0.001	Ea→A	SYN	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1i	0.314	A→Ma	SYN	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	0.566	Ma→A	SYNACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	0.818	A→Ea	SYNACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1i	1.131	Ea→A	ACK	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1i	1.132	Ea→A	DATA (RCV+TR)	2.26	0.0441	0.0441	0.0863	5.5e-05	6.6e-05
S1i	1.445	A→Ma	ACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	1.445	A→Ma	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	1.697	Ma→A	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	1.697	Ma→A	DATA (RCV+TR)	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	1.949	A→Ea	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1i	1.950	A→Ea	DATA (RCV+TR)	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1i	2.263	Ea→A	DACK	2.25	0.0441	0.0441	0.0862	5.5e-05	6.5e-05
S1i	2.576	A→Ma	DACK	1.82	0.0357	0.0357	0.0701	2.4e-05	3.0e-05
S1i	50.501	Ma→A	DATA (2×RCV+2×TR)	1.83	0.0359	0.0359	0.0706	2.5e-05	3.1e-05
S1i	50.755	A→Ea	DATA (2×RCV+2×TR)	2.23	0.0436	0.0436	0.0853	5.3e-05	6.3e-05
S1i	51.065	Ea→A	DACK	2.23	0.0436	0.0436	0.0853	5.3e-05	6.3e-05
S1i	51.065	Ea→A	DATA (2×RCV+2×TR)	2.23	0.0436	0.0436	0.0853	5.3e-05	6.3e-05
S1i	51.374	A→Ma	DACK	1.83	0.0359	0.0359	0.0706	2.5e-05	3.1e-05
S1i	51.375	A→Ma	DATA (2×RCV+2×TR)	1.83	0.0359	0.0359	0.0706	2.5e-05	3.1e-05
S1i	51.629	Ma→A	DACK	1.83	0.0359	0.0359	0.0706	2.5e-05	3.1e-05
S1i	51.883	A→Ea	DACK	2.23	0.0436	0.0436	0.0853	5.3e-05	6.3e-05

E3. Balances and capital

Conservation check. After every ledger event the simulator checks, for each asset, that the opening supply equals the sum of available balances, order holds, contract holds, escrow, and transfers that have been sent but not yet imported. It also checks that every balance is non-negative and that every buy hold equals the remaining quantity times the limit price. For each share registry, it checks that the custody recorded for each exchange equals the shares held there plus the shares in transit to and from it. No run ever failed a check. Table A30 gives the final position of every account that changed.

Table A30: Final balances. Accounts not listed end with their opening balances, and all other balances are zero.

Run	Final balances of the accounts that changed
S1a	Alice at Earth: \$50,000, 1,000 Ares; Bob at Mars: \$150,000, 2,000 Ares
S1b	Alice at Earth: \$50,000, 1,000 Belt; Cara at Ceres: \$200,000
S1c	Fin at Earth: \$90,000, 600 Terra; Eve at Uranus: \$10,000, 400 Terra
S1d	Alice at Earth: \$200,000; Cara at Ceres: \$50,000, 1,000 Belt
S1e	Alice at Earth: \$100,000; Cara at Ceres: \$150,000, 1,000 Belt
S1g	No change
S1i	Alice at Earth: \$149,040, 10 Ares; Bob at Mars: \$50,960, 2,990 Ares

Ledger at the peak of the futures. Between hour 2.502 and hour 326.502 of S1d, Alice’s \$80,000 and Cara’s \$80,000 sit in contract escrow at Mars. Each amount backs the maximum loss of that side (40Q) and is released only by the fixing. All other assets are free: Alice has \$70,000 at Earth, Cara has \$20,000 and 1,000 Belt at Ceres, Bob has \$50,000 and 3,000 Ares at Mars, Dax has \$100,000 at Neptune, Eve has

\$50,000 at Uranus, and Fin has \$50,000 and 1,000 Terra at Earth. Every dollar therefore has one owner, one location and one use.

Table A31 shows when each settled position stops being an obligation, when it is backed by a funded transfer, and when it is spendable at home.

Table A31: When each settled position is discharged, backed by a funded transfer, and spendable at home (hours, no loss).

Position	Obligation discharged	Backed claim exists	Spendable at home
S1a, Alice's shares	Fill at 2.502	Fill at 2.502	3.067
S1a, Bob's cash	Fill at 2.502	— (local)	2.502
S1d and S1e payouts	326.502	326.502	Cara 328.032, Alice 328.174

E3+. State histories and settlement moments

Table A32 is the opening balance sheet. For each scenario the next tables list every change in the positions of its two parties, one row per account and asset that changed, so that the whole ledger history can be read off without the trace files. “Home” is available at the home exchange, “At host” is available in the remote trading account, “Locked” is order and instruction holds plus contract margin, and “Transit” is value debited at one exchange and not yet credited at another. All other accounts are unchanged in every run.

Table A32: Opening balance sheet at hour 0. Every balance is available at the account’s home exchange; remote trading accounts, holds, escrow and transit are all zero. The share registries record the same custody: Mars 3,000 Ares at Mars, Ceres 1,000 Belt at Ceres, Earth 1,000 Terra at Earth.

Account	Home exchange	NeoDollars	Ares	Belt	Terra
Alice	Earth	\$150,000	—	—	—
Bob	Mars	\$50,000	3,000	—	—
Cara	Ceres	\$100,000	—	1,000	—
Dax	Neptune	\$100,000	—	—	—
Eve	Uranus	\$50,000	—	—	—
Fin	Earth	\$50,000	—	—	1,000
Total	5 settlements	\$500,000	3,000	1,000	1,000
Each exchange (own account)	all 9	\$0	—	—	—

Table A33: State history of S1a: Alice (Earth) buys 1,000 Ares from Bob (Mars) at \$100.

h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003	Order admitted EQ-sell	Bob	Ares	2,000	0	1,000	0
0.0003	Export EA-X1 (fund)	Alice	ND	\$50,000	0	0	\$100,000
1.6966	Import EA-X1 (fund)	Alice	ND	\$50,000	\$100,000	0	0
2.5024	Order admitted EQ-buy	Alice	ND	\$50,000	0	\$100,000	0
2.5024	Fill 1,000 Ares @ \$100	Alice	ND	\$50,000	0	0	0
2.5024	Fill 1,000 Ares @ \$100	Alice	Ares	0	1,000	0	0
2.5024	Fill 1,000 Ares @ \$100	Bob	ND	\$150,000	0	0	0
2.5024	Fill 1,000 Ares @ \$100	Bob	Ares	2,000	0	0	0
2.5024	Export MA-X1 (auto fill)	Alice	Ares	0	0	0	1,000
3.0674	Import MA-X1 (auto fill)	Alice	Ares	1,000	0	0	0

Table A34: State history of S1b: Alice (Earth) buys 1,000 Belt from Cara (Ceres) at \$100.

h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003	Order admitted EQB-sell	Cara	Belt	0	0	1,000	0
0.0003	Export EA-X1 (fund)	Alice	ND	\$50,000	0	0	\$100,000
1.6697	Import EA-X1 (fund)	Alice	ND	\$50,000	\$100,000	0	0
2.6085	Order admitted EQB-buy	Alice	ND	\$50,000	0	\$100,000	0
2.6085	Fill 1,000 Belt @ \$100	Alice	ND	\$50,000	0	0	0
2.6085	Fill 1,000 Belt @ \$100	Alice	Belt	0	1,000	0	0
2.6085	Fill 1,000 Belt @ \$100	Cara	ND	\$200,000	0	0	0
2.6085	Fill 1,000 Belt @ \$100	Cara	Belt	0	0	0	0
2.6085	Export CE-X1 (auto fill)	Alice	Belt	0	0	0	1,000
3.1644	Import CE-X1 (auto fill)	Alice	Belt	1,000	0	0	0

Table A35: State history of S1c: Eve (Uranus) buys 400 Terra from Fin (Earth) at \$100.

h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003	Order admitted EQT-sell	Fin	Terra	600	0	400	0
0.0003	Export UR-X1 (fund)	Eve	ND	\$10,000	0	0	\$40,000
7.9134	Import UR-X1 (fund)	Eve	ND	\$10,000	\$40,000	0	0

(continued)

	h	Event	Account	Asset	Home	At host	Locked	Transit
13.1878		Order admitted EQT-buy	Eve	ND	\$10,000	0	\$40,000	0
13.1878		Fill 400 Terra @ \$100	Eve	ND	\$10,000	0	0	0
13.1878		Fill 400 Terra @ \$100	Eve	Terra	0	400	0	0
13.1878		Fill 400 Terra @ \$100	Fin	ND	\$90,000	0	0	0
13.1878		Fill 400 Terra @ \$100	Fin	Terra	600	0	0	0
13.1878		Export EA-X1 (auto fill)	Eve	Terra	0	0	0	400
15.8244		Import EA-X1 (auto fill)	Eve	Terra	400	0	0	0

Table A36: State history of S1d: capped futures, rising path, $Q = 2,000$ (Alice long, Cara short).

	h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003		Export EA-X1 (fund)	Alice	ND	\$70,000	0	0	\$80,000
0.0003		Export CE-X1 (fund)	Cara	ND	\$20,000	0	0	\$80,000
1.4868		Import CE-X1 (fund)	Cara	ND	\$20,000	\$80,000	0	0
1.6966		Import EA-X1 (fund)	Alice	ND	\$70,000	\$80,000	0	0
2.1410		Margin held F1	Cara	ND	\$20,000	0	\$80,000	0
2.5024		Margin held F1	Alice	ND	\$70,000	0	\$80,000	0
326.5024		Contract fixed F1	Alice	ND	\$70,000	\$130,000	0	0
326.5024		Contract fixed F1	Cara	ND	\$20,000	\$30,000	0	0
326.5024		Export MA-X1 (payout)	Alice	ND	\$70,000	0	0	\$130,000
326.5024		Export MA-X2 (payout)	Cara	ND	\$20,000	0	0	\$30,000
328.0321		Import MA-X2 (payout)	Cara	ND	\$50,000	0	0	0
328.1736		Import MA-X1 (payout)	Alice	ND	\$200,000	0	0	0

Table A37: State history of S1e: capped futures, falling path, $Q = 2,000$.

	h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003		Export EA-X1 (fund)	Alice	ND	\$70,000	0	0	\$80,000
0.0003		Export CE-X1 (fund)	Cara	ND	\$20,000	0	0	\$80,000
1.4868		Import CE-X1 (fund)	Cara	ND	\$20,000	\$80,000	0	0
1.6966		Import EA-X1 (fund)	Alice	ND	\$70,000	\$80,000	0	0
2.1410		Margin held F1	Cara	ND	\$20,000	0	\$80,000	0
2.5024		Margin held F1	Alice	ND	\$70,000	0	\$80,000	0
326.5024		Contract fixed F1	Alice	ND	\$70,000	\$30,000	0	0
326.5024		Contract fixed F1	Cara	ND	\$20,000	\$130,000	0	0
326.5024		Export MA-X1 (payout)	Alice	ND	\$70,000	0	0	\$30,000
326.5024		Export MA-X2 (payout)	Cara	ND	\$20,000	0	0	\$130,000
328.0321		Import MA-X2 (payout)	Cara	ND	\$150,000	0	0	0
328.1736		Import MA-X1 (payout)	Alice	ND	\$100,000	0	0	0

Table A38: State history of S1f: capped futures, boundary case $Q = 2,500$.

	h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003		Export EA-X1 (fund)	Alice	ND	\$50,000	0	0	\$100,000
0.0003		Export CE-X1 (fund)	Cara	ND	0	0	0	\$100,000
1.4868		Import CE-X1 (fund)	Cara	ND	0	\$100,000	0	0
1.6966		Import EA-X1 (fund)	Alice	ND	\$50,000	\$100,000	0	0
2.1410		Margin held F1	Cara	ND	0	0	\$100,000	0
2.5024		Margin held F1	Alice	ND	\$50,000	0	\$100,000	0
326.5024		Contract fixed F1	Alice	ND	\$50,000	\$162,500	0	0
326.5024		Contract fixed F1	Cara	ND	0	\$37,500	0	0
326.5024		Export MA-X1 (payout)	Alice	ND	\$50,000	0	0	\$162,500

(continued)

	h	Event	Account	Asset	Home	At host	Locked	Transit
326.5024		Export MA-X2 (payout)	Cara	ND	0	0	0	\$37,500
328.0321		Import MA-X2 (payout)	Cara	ND	\$37,500	0	0	0
328.1736		Import MA-X1 (payout)	Alice	ND	\$212,500	0	0	0

Table A39: State history of S1g: capped futures, $Q = 2,501$ (Cara cannot fund; Alice is refunded).

	h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003		Export EA-X1 (fund)	Alice	ND	\$49,960	0	0	\$100,040
1.6966		Import EA-X1 (fund)	Alice	ND	\$49,960	\$100,040	0	0
2.5024		Margin held F1	Alice	ND	\$49,960	0	\$100,040	0
168.0000		Margin released F1	Alice	ND	\$49,960	\$100,040	0	0
169.2336		Export MA-X1 (withdraw)	Alice	ND	\$49,960	0	0	\$100,040
169.7939		Import MA-X1 (withdraw)	Alice	ND	\$150,000	0	0	0

Table A40: State history of S1i: repeat trading on a \$10,000 Mars account.

	h	Event	Account	Asset	Home	At host	Locked	Transit
0.0003		Export EA-X1 (fund)	Alice	ND	\$140,000	0	0	\$10,000
0.0003		Order admitted RT-s1	Bob	Ares	2,990	0	10	0
0.0003		Order admitted RT-s2	Bob	Ares	2,985	0	15	0
1.6966		Import EA-X1 (fund)	Alice	ND	\$140,000	\$10,000	0	0
2.5024		Order admitted RT-o1	Alice	ND	\$140,000	\$6,000	\$4,000	0
2.5024		Fill 10 Ares @ \$97	Alice	ND	\$140,000	\$6,030	\$3,000	0
2.5024		Fill 10 Ares @ \$97	Alice	Ares	0	10	0	0
2.5024		Fill 10 Ares @ \$97	Bob	ND	\$50,970	0	0	0
2.5024		Fill 10 Ares @ \$97	Bob	Ares	2,985	0	5	0
2.5024		Fill 5 Ares @ \$100	Alice	ND	\$140,000	\$6,030	\$2,500	0
2.5024		Fill 5 Ares @ \$100	Alice	Ares	0	15	0	0
2.5024		Fill 5 Ares @ \$100	Bob	ND	\$51,470	0	0	0
2.5024		Fill 5 Ares @ \$100	Bob	Ares	2,985	0	0	0
2.5024		Order admitted RT-o2	Alice	ND	\$140,000	\$2,030	\$6,500	0
8.5021		Cancelled RT-o2	Alice	ND	\$140,000	\$6,030	\$2,500	0
14.5019		Order admitted RT-o3	Alice	Ares	0	5	10	0
15.2627		Order admitted RT-b1	Bob	ND	\$50,460	0	\$1,010	0
15.2627		Fill 10 Ares @ \$101	Alice	ND	\$140,000	\$7,040	\$2,500	0
15.2627		Fill 10 Ares @ \$101	Alice	Ares	0	5	0	0
15.2627		Fill 10 Ares @ \$101	Bob	ND	\$50,460	0	0	0
15.2627		Fill 10 Ares @ \$101	Bob	Ares	2,995	0	0	0
26.3627		Order admitted RT-s3	Bob	Ares	2,990	0	5	0
26.3627		Fill 5 Ares @ \$100	Alice	ND	\$140,000	\$7,040	\$2,000	0
26.3627		Fill 5 Ares @ \$100	Alice	Ares	0	10	0	0
26.3627		Fill 5 Ares @ \$100	Bob	ND	\$50,960	0	0	0
26.3627		Fill 5 Ares @ \$100	Bob	Ares	2,990	0	0	0
26.5015		Cancelled RT-o1	Alice	ND	\$140,000	\$9,040	0	0
50.5006		Export MA-X1 (withdraw)	Alice	ND	\$140,000	0	0	\$9,040
50.5006		Export MA-X2 (withdraw)	Alice	Ares	0	0	0	10
51.0642		Import MA-X1 (withdraw)	Alice	ND	\$149,040	0	0	0
51.0642		Import MA-X2 (withdraw)	Alice	Ares	10	0	0	0

Settlement moments. Table A41 lists every obligation in every scenario. An obligation is *discharged* when the duty it records is met (a fill, a fixing, or a transfer credited at its destination); a claim is *backed* once the value has been debited into a transfer, so nobody can spend it twice; and it is *spendable* when the

owner can use it, either at the host or at home.

Table A41: Settlement moments of every obligation in every S1 scenario (hours, no loss).

Run	Obligation	Created h	Discharged h	Backed	Spendable
S1a	Fund \$100,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1a	Fill 1,000 Ares @ \$100: Alice from Bob	2.502	2.502	local	2.502 at host
S1a	Auto Fill 1,000 Ares for Alice, Ma→Ea	2.502	3.067	2.502	3.067 at home
S1b	Fund \$100,000 for Alice, Ea→Ce	0.000	1.670	0.000	1.670 at host
S1b	Fill 1,000 Belt @ \$100: Alice from Cara	2.608	2.608	local	2.608 at host
S1b	Auto Fill 1,000 Belt for Alice, Ce→Ea	2.608	3.164	2.608	3.164 at home
S1c	Fund \$40,000 for Eve, Ur→Ea	0.000	7.913	0.000	7.913 at host
S1c	Fill 400 Terra @ \$100: Eve from Fin	13.188	13.188	local	13.188 at host
S1c	Auto Fill 400 Terra for Eve, Ea→Ur	13.188	15.824	13.188	15.824 at home
S1d	Fund \$80,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1d	Fund \$80,000 for Cara, Ce→Ma	0.000	1.487	0.000	1.487 at host
S1d	Contract F1 (escrow \$160,000)	2.502	at fixing	escrow from 2.502	—
S1d	Fixing of F1 at 125	326.502	326.502	326.502 (payouts)	—
S1d	Payout \$130,000 for Alice, Ma→Ea	326.502	328.174	326.502	328.174 at home
S1d	Payout \$30,000 for Cara, Ma→Ce	326.502	328.032	326.502	328.032 at home
S1e	Fund \$80,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1e	Fund \$80,000 for Cara, Ce→Ma	0.000	1.487	0.000	1.487 at host
S1e	Contract F1 (escrow \$160,000)	2.502	at fixing	escrow from 2.502	—
S1e	Fixing of F1 at 75	326.502	326.502	326.502 (payouts)	—
S1e	Payout \$30,000 for Alice, Ma→Ea	326.502	328.174	326.502	328.174 at home
S1e	Payout \$130,000 for Cara, Ma→Ce	326.502	328.032	326.502	328.032 at home
S1f	Fund \$100,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1f	Fund \$100,000 for Cara, Ce→Ma	0.000	1.487	0.000	1.487 at host
S1f	Contract F1 (escrow \$200,000)	2.502	at fixing	escrow from 2.502	—
S1f	Fixing of F1 at 125	326.502	326.502	326.502 (payouts)	—
S1f	Payout \$162,500 for Alice, Ma→Ea	326.502	328.174	326.502	328.174 at home
S1f	Payout \$37,500 for Cara, Ma→Ce	326.502	328.032	326.502	328.032 at home
S1g	Fund \$100,040 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1g	Withdraw \$100,040 for Alice, Ma→Ea	169.234	169.794	169.234	169.794 at home
S1i	Fund \$10,000 for Alice, Ea→Ma	0.000	1.697	0.000	1.697 at host
S1i	Fill 10 Ares @ \$97: Alice from Bob	2.502	2.502	local	2.502 at host
S1i	Fill 5 Ares @ \$100: Alice from Bob	2.502	2.502	local	2.502 at host
S1i	Fill 10 Ares @ \$101: Bob from Alice	15.263	15.263	local	15.263 at host
S1i	Fill 5 Ares @ \$100: Alice from Bob	26.363	26.363	local	26.363 at host
S1i	Withdraw \$9,040 for Alice, Ma→Ea	50.501	51.064	50.501	51.064 at home

(continued)

Run	Obligation	Created h	Discharged h	Backed	Spendable
S1i	Withdraw 10 Ares for Alice, Ma→Ea	50.501	51.064	50.501	51.064 at home

S2. Stress and recovery

Choice of incident. We ran the futures scenario on both price paths under every allowed incident on a grid of start times between hour 0 and hour 420. The grid covered a 72-hour isolation of each settlement starting every hour, a 6-hour forced loss at each of the 11 nodes starting every 0.5 hours, and an endpoint reset of each exchange every 0.25 hours, 56,280 runs in total. We measured damage as the extra hours until both payouts are spendable, with a penalty of 10^6 if the contract never opens. Table A42 gives the worst start time for each kind of incident and node, and Figure A2 shows the damage against start time.

Table A42: Worst start time for each kind of incident and node, over both price paths.

Incident	Node	Start h	Opened h	Done h	Extra h
reset	Mars	0.75	128.13	453.79	+125.6
reset	Earth	327.25	2.50	453.73	+125.6
reset	Ceres	327.25	2.50	453.16	+125.0
isolation	Earth	1	77.88	403.55	+75.4
isolation	Mars	1	77.88	403.55	+75.4
isolation	Ceres	310	2.50	403.11	+74.9



Figure A2: Damage (extra hours until both payouts are home, capped at 200) against incident start time on the rising path. Coloured lines are the three exchanges in the scenario; grey lines are the other nodes.

We chose the endpoint reset of the Mars exchange at hour 0.75, which is the worst incident on both price paths. It does more damage than the obvious choice, isolating the host at payout time (about 75 hours), even though no party can observe it. At hour 0.75, Mars has just answered Alice’s and Cara’s handshakes. The reset erases its session state, but its SYN-ACK replies are already in flight. Earth and Ceres therefore believe that the sessions are open and send their funding transfers into sessions that Mars no longer knows. As the brief requires, Mars ignores them. The senders learn nothing until four endpoint attempts of $R_e \approx 25$ hours have expired. They then wait for the 24-hour application backoff and resubmit. The courier log of the rising path shows this sequence. Earth and Ceres open their sessions with Mars at hour 0.0003. Ceres declares its delivery unknown at hour 100.97 and Earth at hour 101.64. Each schedules an application resubmission 24 hours later, and they resubmit in new sessions at hours 124.97 and 125.64.

State of every obligation during the incident. Alice’s and Cara’s \$80,000 have been debited at home and are in transit. They still belong to Alice and Cara but cannot be spent anywhere, for about 125 hours. No contract exists yet, so nobody owes anything and neither margin can be lost. Bob’s shares and all other accounts are not affected. Local service at Mars never stops, because the reset only erased transport state.

Recovery. Recovery here means that financial service resumes, not only that a packet gets through. The transfers are imported at Mars, both margins are held and the contract opens at hour 128.13. This is before the 168-hour deadline, so the trade still takes place with the full 300-hour term, and both payouts arrive

home. Table A43 compares the variants. Across the search, isolation and forced loss matter only around the opening (hours 0 to 3) and in the payout window (hours 255 to 330), and a reset matters only if it falls between a handshake and the first data packet.

Table A43: The chosen incident under each variant, without random loss. The falling price path gives identical results.

Variant	Opened h	Done h	Packets	Originations
No incident	2.50	328.17	141	15
Incident, baseline design	128.13	453.79	197	19
Incident, no maintenance	128.13	453.79	197	19
No incident, no maintenance	2.50	328.17	141	15
Incident with restart notice (A1)	3.82	329.49	189	21
Incident with no backoff	104.13	429.80	197	19

Maintenance. Removing the two scheduled maintenance windows makes no difference to this incident. Neither window (Relay B to Neptune from hour 2 to 26, and Relay B to Ceres from hour 240 to 264) lies on the routes that the runs use, which are Earth, Relay A, Mars and Ceres, Relay A, Mars. The timing is set by the geometry alone.

Alternative A1. This alternative changes one rule and keeps the same funding and guarantees. After a reset, an exchange sends a RESTART notice to every peer it has had a session with in the last 7 days. The notice is one 64-byte backbone record and counts against the sender’s quota. A peer that receives it abandons its old sessions with that exchange and immediately resubmits all unacknowledged records. With A1 the contract opens at hour 3.82 instead of hour 128.13, and fewer packets are needed. Removing the backoff alone, as a sensitivity test, saves only 24 hours. A1 is not part of the evaluated design v3; it is the first change for the next revision. Table A44 gives the results with random loss.

Table A44: The chosen incident with random loss (1,000 runs each).

Variant	Done	Median h	p90 h	p99 h	P(open)	Mean pkts
incident	1000/1000	455.33	457.27	460.61	100.0%	206
incident restart notice	1000/1000	331.07	333.68	337.66	100.0%	197

Complete trace of the incident. Table A45 lists every event of the selected incident up to the opening; the files `traces/S2_reset_Mars_{launches,transport,ledger}.csv` hold the whole run, including the 300-hour contract and the payouts, which then follow S1d shifted by the delay. The table shows the four steps of the 125-hour penalty: the handshakes complete before the reset, so Earth and Ceres send their transfers into sessions Mars has forgotten; Mars ignores every copy; each sender’s four endpoint attempts expire (delivery unknown); and after the 24-hour application backoff each resubmits on a new session, which succeeds within about two hours.

Table A45: Event-by-event trace of the S2 incident (rising path, no loss) from hour 0 to the opening at 128.126: every backbone launch except the 66 hop receipts, every direct copy, every courier event and every ledger event. Ea = Earth, Ma = Mars, Ce = Ceres, A = Relay A.

h	At	Event	Detail
0.000	Earth	session open	session s26 to Mars via Earth, Relay A, Mars
0.000	Ceres	session open	session s27 to Mars via Ceres, Relay A, Mars
0.000	Alice	local instruction	FUND \$80,000 to Mars
0.000	Earth	export	EA-X1: \$80,000 of Alice debited, in transit to Mars
0.000	Cara	local instruction	FUND \$80,000 to Mars
0.000	Ceres	export	CE-X1: \$80,000 of Cara debited, in transit to Mars
0.001	Ce→A	launch	SYN s27, arrives 0.243
0.001	Ea→A	launch	SYN s26, arrives 0.313
0.244	A→Ma	launch	SYN s27, arrives 0.495
0.314	A→Ma	launch	SYN s26, arrives 0.565
0.496	Ma→A	launch	SYNACK s27, arrives 0.748

(continued)

h	At	Event	Detail
0.566	Ma→A	launch	SYNACK s26, arrives 0.818
0.748	A→Ce	launch	SYNACK s27, arrives 0.991
0.750	Mars	endpoint reset	all session state lost; ledger and durable outbox kept
0.818	A→Ea	launch	SYNACK s26, arrives 1.131
0.992	Ce→A	launch	ACK s27, arrives 1.234
0.992	Ce→A	launch	DATA s27 (carries DATA), arrives 1.235
1.131	Ea→A	launch	ACK s26, arrives 1.444
1.132	Ea→A	launch	DATA s26 (carries DATA), arrives 1.444
1.235	A→Ma	launch	ACK s27, arrives 1.487
1.235	A→Ma	launch	DATA s27 (carries DATA), arrives 1.487
1.445	A→Ma	launch	ACK s26, arrives 1.696
1.445	A→Ma	launch	DATA s26 (carries DATA), arrives 1.697
1.487	Mars	ignored	ACK on s27 from Ceres: Mars lost this session in the reset
1.487	Mars	ignored	DATA on s27 from Ceres: Mars lost this session in the reset
1.696	Mars	ignored	ACK on s26 from Earth: Mars lost this session in the reset
1.697	Mars	ignored	DATA on s26 from Earth: Mars lost this session in the reset
25.982	Ce→A	launch	DATA s27 (carries DATA), arrives 26.225
26.225	A→Ma	launch	DATA s27 (carries DATA), arrives 26.478
26.261	Ea→A	launch	DATA s26 (carries DATA), arrives 26.572
26.478	Mars	ignored	DATA on s27 from Ceres: Mars lost this session in the reset
26.573	A→Ma	launch	DATA s26 (carries DATA), arrives 26.826
26.826	Mars	ignored	DATA on s26 from Earth: Mars lost this session in the reset
50.974	Ce→A	launch	DATA s27 (carries DATA), arrives 51.217
51.218	A→Ma	launch	DATA s27 (carries DATA), arrives 51.472
51.390	Ea→A	launch	DATA s26 (carries DATA), arrives 51.699
51.472	Mars	ignored	DATA on s27 from Ceres: Mars lost this session in the reset
51.700	A→Ma	launch	DATA s26 (carries DATA), arrives 51.954
51.954	Mars	ignored	DATA on s26 from Earth: Mars lost this session in the reset
75.969	Ce→A	launch	DATA s27 (carries DATA), arrives 76.212
76.213	A→Ma	launch	DATA s27 (carries DATA), arrives 76.467
76.467	Mars	ignored	DATA on s27 from Ceres: Mars lost this session in the reset
76.518	Ea→A	launch	DATA s26 (carries DATA), arrives 76.825
76.826	A→Ma	launch	DATA s26 (carries DATA), arrives 77.080
77.080	Mars	ignored	DATA on s26 from Earth: Mars lost this session in the reset
100.966	Ceres	delivery unknown	s27 to Mars: four endpoint attempts unacknowledged
100.966	Ceres	app resubmit scheduled	resubmission of the transfer scheduled for 124.966
101.643	Earth	delivery unknown	s26 to Mars: four endpoint attempts unacknowledged
101.643	Earth	app resubmit scheduled	resubmission of the transfer scheduled for 125.643
124.966	Ceres	app resubmit	transfer resubmitted to Mars on a new session
124.966	Ceres	session open	session s28 to Mars via Ceres, Relay A, Mars
124.966	Ce→A	launch	SYN s28, arrives 125.209
125.210	A→Ma	launch	SYN s28, arrives 125.467
125.467	Ma→A	launch	SYNACK s28, arrives 125.724

(continued)

	h	At	Event	Detail
125.643	Earth		app resubmit	transfer resubmitted to Mars on a new session
125.643	Earth		session open	session s29 to Mars via Earth, Relay A, Mars
125.643	Ea→A		launch	SYN s29, arrives 125.948
125.724	A→Ce		launch	SYNACK s28, arrives 125.967
125.948	A→Ma		launch	SYN s29, arrives 126.205
125.968	Ce→A		launch	ACK s28, arrives 126.211
125.968	Ce→A		launch	DATA s28 (carries DATA), arrives 126.211
126.205	Ma→A		launch	SYNACK s29, arrives 126.462
126.212	A→Ma		launch	ACK s28, arrives 126.468
126.212	A→Ma		launch	DATA s28 (carries DATA), arrives 126.469
126.463	A→Ea		launch	SYNACK s29, arrives 126.767
126.469	Mars		import	CE-X1: \$80,000 credited to Cara
126.469	Mars		received queued	RECEIVED for CE-X1 queued to Ceres
126.469	Ma→A		launch	DACK s28, arrives 126.726
126.470	Ma→A		launch	DATA s28 (carries DATA), arrives 126.726
126.726	A→Ce		launch	DACK s28, arrives 126.970
126.727	A→Ce		launch	DATA s28 (carries DATA), arrives 126.970
126.767	Ea→A		launch	ACK s29, arrives 127.071
126.767	Ea→A		launch	DATA s29 (carries DATA), arrives 127.071
126.970	Ceres		receipt known	CE-X1 resolved
126.970	Ce→A		launch	DACK s28, arrives 127.214
126.971	Ce→Ma		launch	direct CINSTR copy 1 from Cara, arrives 127.127
126.987	Ce→Ma		launch	direct CINSTR copy 2 from Cara, arrives 127.143
127.072	A→Ma		launch	ACK s29, arrives 127.328
127.072	A→Ma		launch	DATA s29 (carries DATA), arrives 127.329
127.127	Mars		margin held	Cara: \$80,000 margin held
127.127	Mars		status queued	STATUS record to Ceres
127.127	Ma→A		launch	DATA s28 (carries DATA), arrives 127.384
127.143	Mars		duplicate cinstr	later copy from Cara ignored
127.214	A→Ma		launch	DACK s28, arrives 127.471
127.329	Mars		import	EA-X1: \$80,000 credited to Alice
127.329	Mars		received queued	RECEIVED for EA-X1 queued to Earth
127.329	Ma→A		launch	DACK s29, arrives 127.586
127.329	Ma→A		launch	DATA s29 (carries DATA), arrives 127.586
127.384	A→Ce		launch	DATA s28 (carries DATA), arrives 127.627
127.586	A→Ea		launch	DACK s29, arrives 127.890
127.587	A→Ea		launch	DATA s29 (carries DATA), arrives 127.891
127.627	Ceres		status received	Cara told HELD
127.628	Ce→A		launch	DACK s28, arrives 127.871
127.872	A→Ma		launch	DACK s28, arrives 128.128
127.891	Earth		receipt known	EA-X1 resolved
127.891	Ea→A		launch	DACK s29, arrives 128.195
127.891	Ea→Ma		launch	direct CINSTR copy 1 from Alice, arrives 128.126
127.908	Ea→Ma		launch	direct CINSTR copy 2 from Alice, arrives 128.143
127.925	Ea→Ma		launch	direct CINSTR copy 3 from Alice, arrives 128.159
128.126	Mars		margin held	Alice: \$80,000 margin held
128.126	Mars		contract open	F1 opens, escrow \$160,000
128.126	Mars		status queued	STATUS record to Earth
128.126	Mars		status queued	STATUS record to Ceres
128.126	Ma→A		launch	DATA s29 (carries DATA), arrives 128.383
128.127	Ma→A		launch	DATA s28 (carries DATA), arrives 128.383
128.143	Mars		duplicate cinstr	later copy from Alice ignored

(continued)

	h	At	Event	Detail
128.159	Mars		duplicate cinstr	later copy from Alice ignored
128.196	A→Ma		launch	DACK s29, arrives 128.452
128.384	A→Ea		launch	DATA s29 (carries DATA), arrives 128.688
128.384	A→Ce		launch	DATA s28 (carries DATA), arrives 128.627
128.627	Ceres		status received	Cara told OPEN
128.628	Ce→A		launch	DACK s28, arrives 128.871
128.688	Earth		status received	Alice told OPEN
128.688	Ea→A		launch	DACK s29, arrives 128.992
128.871	A→Ma		launch	DACK s28, arrives 129.128
128.992	A→Ma		launch	DACK s29, arrives 129.249

S3. Access at every settlement

Funding, withdrawals and payouts travel over the backbone between the client's home exchange and the host; orders and contract instructions travel by direct service from the client to the host. The hosts are Mars (Ares and the futures), Ceres (Belt) and Earth (Terra). Tables A46 and A47 show access by each service. Availability is sampled every minute with maintenance applied. The only availability changes between hour 0 and hour 300 are on Mercury's direct paths: to Mars it is open for 95% of the first day and closed all of day 300 (hours 22.5 to 357), and to Ceres it is open for 37% of day 300. R8 forwards around both.

Table A46: Backbone access from each settlement to the three hosts at hours 0 and 300: relay or relays used (A, B) and one-way delay there and back in minutes. Availability over the next 24 hours is 100% unless shown.

From	Mars, h 0	Mars, h 300	Ceres, h 0	Ceres, h 300	Earth, h 0	Earth, h 300
Mercury	A 42/42	A 42/42	A 42/42	A 41/41	A 46/46	A 44/44
Venus	A 37/37	A 36/36	A 36/36	A 35/35	A 41/41	A 38/38
Earth	A 34/34	A 33/33	A 33/33	A 32/32	local	local
Mars	local	local	A 30/30	A 30/31	A 34/34	A 33/33
Ceres	A 30/30	A 31/30	local	local	A 33/33	A 32/32
Jupiter	B 40/40	B 39/39	B 41/41	B 41/41	B 51/51	B 50/50
Saturn	A 76/76	A 77/77	A 75/75	A 76/76	A 79/79	A 79/79
Uranus	A 155/155	A 155/155	A 154/154	A 154/154	A 158/158	A 157/157
Neptune	A 247/247	A 248/248	A 246/246	A 247/247	A 250/250	A 250/250

Table A47: Direct access from each settlement to the three hosts at hours 0 and 300: one-way delay in minutes, availability over the next 24 hours, and the number of copies the client sends.

From	Mars, h 0	Mars, h 300	Ceres, h 0	Ceres, h 300	Earth, h 0	Earth, h 300
Mercury	17, 95%, 3	17, 0%, 3	26, 100%, 4	26, 37%, 4	11, 100%, 2	9, 100%, 2
Venus	15, 100%, 3	14, 100%, 3	24, 100%, 3	23, 100%, 3	3, 100%, 2	3, 100%, 2
Earth	14, 100%, 3	14, 100%, 3	23, 100%, 3	21, 100%, 3	local	local
Mars	local	local	9, 100%, 2	9, 100%, 2	14, 100%, 3	14, 100%, 3
Ceres	9, 100%, 2	9, 100%, 2	local	local	23, 100%, 3	21, 100%, 3
Jupiter	37, 100%, 4	36, 100%, 4	34, 100%, 4	33, 100%, 4	50, 100%, 5	49, 100%, 5
Saturn	75, 100%, 7	76, 100%, 8	75, 100%, 7	75, 100%, 7	70, 100%, 7	70, 100%, 7
Uranus	150, 100%, 8	150, 100%, 8	141, 100%, 8	141, 100%, 8	158, 100%, 8	157, 100%, 8
Neptune	246, 100%, 8	248, 100%, 8	245, 100%, 8	246, 100%, 8	240, 100%, 8	240, 100%, 8

Relocation. We re-ran the same client, a \$100,000 Ares purchase with home delivery from a comparable funded account, with its home moved to each settlement in turn (Table A48). A Mars home is best overall (the purchase is local, with no packets) and is left out of the ranking. Table A49 gives the step-by-step timeline for the best (Ceres), median (Mercury) and worst (Neptune) homes. The sequence is the same everywhere; only the backbone and direct delays and the number of direct copies change.

Table A48: The same \$100,000 Ares purchase with the client moved to each settlement, ranked by completion time without loss. Times in hours; P(t) is the probability of completing within t hours (1,000 lossy runs each).

Home	No loss	Pkts	Median	p90	p99	P(24)	P(72)	P(168)	Mean pkts
Mars	0.00	0	0.00	0.00	0.00	1.00	1.00	1.00	0
Ceres	2.64	46	2.64	4.14	7.11	1.00	1.00	1.00	48
Earth	3.07	47	3.07	4.69	6.32	1.00	1.00	1.00	50
Venus	3.33	47	3.35	6.34	8.28	1.00	1.00	1.00	50
Mercury	3.81	47	3.83	7.22	9.53	1.00	1.00	1.00	51
Jupiter	3.95	48	3.98	7.27	12.29	1.00	1.00	1.00	51
Saturn	7.57	51	10.59	15.15	21.63	0.99	1.00	1.00	58
Uranus	15.38	52	26.76	48.49	70.31	0.34	0.99	1.00	68
Neptune	24.68	52	74.43	127.05	201.84	0.00	0.47	0.97	97

Table A49: Step-by-step timeline of the relocated purchase for the best, median and worst homes. The full packet-level traces are in `traces/S3_reloc_<home>_*.csv`.

Step (hours, no loss)	Best: Ceres	Median: Mercury	Worst: Neptune
Funding imported at Mars	1.487	2.119	12.344
RECEIVED known at home	1.983	2.826	16.459
Order sent (direct)	1.983	2.826	16.460
Order arrives and fills	2.141	3.103	20.565
Bob's cash spendable	2.141	3.103	20.565
Shares spendable at home	2.636	3.810	24.681
Order copies $\times$ loss each	$2 \times p$ 0.087	$3 \times p$ 0.147	$8 \times p$ 0.906
Packets (backbone + direct)	44 + 2	44 + 3	44 + 8

Outer settlements in the futures. Table A50 shows the futures with Dax (Neptune) or Eve (Uranus) as the long party against Cara. Eve's position is $Q = 1,250$ so that it fits her \$50,000.

Table A50: Futures with an outer-system long party against Cara. Times in hours; P(open) is the probability that the contract opens before the 168-hour deadline (300 lossy runs each).

Long party	Open, no loss	Done, no loss	P(open)	Median open	Median done
Dax	20.57	356.99	99.7%	62.04	421.46
Eve	12.80	344.56	100.0%	24.13	363.01

Every product at every settlement. Table A51 repeats the relocation for each product. A purchase is local, with no network traffic, only when the client's home is the market's host; every other cell uses the same three steps (funding over the backbone, a direct order, delivery home over the backbone). The time is set by the slower of the client's direct path to the host and the backbone route back, so the columns differ little. The Neptune client completes within 72 hours with probability 0.49 and within 168 hours with probability 0.98.

Of the 10,800 lossy runs, 3 did not trade, all with the client at Neptune (1 Ares, 1 Belt, 1 Futures), and none lost anything. A direct copy from Neptune to Mars is lost with probability about 0.91. In the share runs every copy of the order was lost during its 7-day life (64 copies in 8 sends in the Ares run); it expired and the \$100,000 stayed available to the client at the host. In the futures run the client's instruction did not reach Mars before the 168-hour opening deadline, so the contract was rejected and both margins went back home.

S3 relocations. The same \$100,000 Ares purchase with Alice's account moved to the best, median and worst homes of Table A48.

Table A51: Service by product and settlement. The client’s account is moved to each settlement and buys 1,000 shares of each stock at \$100 at its market with home delivery, or goes long $Q = 2,000$ in the Mars futures against Cara. Share columns: completion hours without loss / median under random loss / probability of completing within 72 hours. Futures column: opening hour without loss / probability of opening before the 168-hour deadline. 300 lossy runs per cell (`products.py`).

Home	Ares at Mars	Belt at Ceres	Terra at Earth	Futures at Mars
Mercury	3.8 / 5.3 / 1.00	3.9 / 4.7 / 1.00	4.0 / 5.6 / 1.00	3.1 / 100.0%
Venus	3.3 / 3.3 / 1.00	3.4 / 3.5 / 1.00	3.4 / 3.5 / 1.00	2.7 / 100.0%
Earth	3.1 / 3.1 / 1.00	3.2 / 3.2 / 1.00	local, 0.0	2.5 / 100.0%
Mars	local, 0.0	2.6 / 2.6 / 1.00	3.1 / 3.1 / 1.00	2.1 / 100.0%
Ceres	2.6 / 2.6 / 1.00	local, 0.0	3.2 / 3.2 / 1.00	2.1 / 100.0%
Jupiter	3.9 / 4.0 / 1.00	4.0 / 4.1 / 1.00	5.1 / 6.8 / 1.00	3.3 / 100.0%
Saturn	7.6 / 10.6 / 1.00	7.5 / 10.5 / 1.00	7.8 / 10.8 / 1.00	6.3 / 100.0%
Uranus	15.4 / 26.8 / 0.99	15.2 / 26.5 / 1.00	15.8 / 27.2 / 0.98	12.8 / 100.0%
Neptune	24.7 / 73.4 / 0.49	24.6 / 73.2 / 0.49	24.9 / 69.4 / 0.52	20.6 / 99.7%

Table A52: Complete trace of the S3 relocation, best home: Alice at Ceres buys 1,000 Ares from Bob (Mars). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Ceres; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Ceres; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Ceres, sent h 0.0000	0.0003	Alice \$150k at Ceres; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Ceres exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ce→Ma, batched with queued records	1.4868	Alice \$150k at Ceres; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Ceres balance	Funds own Mars account: \$100k	Local access at Ceres, sent h 0.0000	0.0003	Alice \$150k at Ceres; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Ceres exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER CE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ce→A h 0.9918, A→Ma h 1.2351	1.4868	Alice \$50k at Ceres, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.4868	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Ce, batched with queued records	1.9826	Alice \$50k at Ceres, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
1.4868	Mars exchange	Authentic new transfer CE-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Ceres (backbone)	1.9826	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.9826	Ceres exchange	MDS from Mars	Posts Ares snapshot to its board (0.50 h old); answers waiting QUOTES	Local access (1 s)	1.9829	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.9826	Ceres exchange	Mars imported CE-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.9829	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 0.50 h old	—	—	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
1.9829	Alice	Published geometry; own funding confirmed	Sends ORDER ×2 copies	Direct Ce→Ma, launches 1.9829, 1.9996; p(loss) 0.087 each, all lost 7.5e-03	2.1410 (first)	Alice \$50k at Ceres, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.1410	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Ceres, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.1410	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Ceres, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home
2.1410	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Ceres	Session from h 0.0006; DATA Ma→A h 2.1412, A→Ce h 2.3935	2.6363	Alice \$50k at Ceres, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.1576	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Ceres, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
2.6363	Ceres exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	3.1321	Alice \$50k at Ceres, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
3.1321	Mars exchange	Ceres imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Ceres, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

Table A53: Complete trace of the S3 relocation, median home: Alice at Mercury buys 1,000 Ares from Bob (Mars). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Mercury; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Mercury; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Mercury, sent h 0.0000	0.0003	Alice \$150k at Mercury; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Mercury exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Me→Ma, batched with queued records	2.1195	Alice \$150k at Mercury; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Mercury balance	Funds own Mars account: \$100k	Local access at Mercury, sent h 0.0000	0.0003	Alice \$150k at Mercury; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Mercury exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER ME-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Me→A h 1.4136, A→Ma h 1.8678	2.1195	Alice \$50k at Mercury, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.1195	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Me, batched with queued records	2.8262	Alice \$50k at Mercury, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.1195	Mars exchange	Authentic new transfer ME-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Mercury (backbone)	2.8262	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.8262	Mercury exchange	MDS from Mars	Posts Ares snapshot to its board (0.71 h old); answers waiting QUOTEs	Local access (1 s)	2.8264	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
2.8262	Mercury exchange	Mars imported ME-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.8264	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 0.71 h old	—	—	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
2.8264	Alice	Published geometry; own funding confirmed	Sends ORDER ×3 copies	Direct Me→Ma, launches 2.8264, 2.8431, 2.8598; p(loss) 0.147 each, all lost 3.2e-03	3.1035 (first)	Alice \$50k at Mercury, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
3.1035	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Mercury, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
3.1035	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Mercury, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home
3.1035	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Mercury	Session from h 0.0006; DATA Ma→A h 3.1037, A→Me h 3.3561	3.8096	Alice \$50k at Mercury, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.1201	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Mercury, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.1368	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Mercury, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
3.8096	Mercury exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	4.5163	Alice \$50k at Mercury, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home
4.5163	Mars exchange	Mercury imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Mercury, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

Table A54: Complete trace of the S3 relocation, worst home: Alice at Neptune buys 1,000 Ares from Bob (Mars). Every ledger event, direct message and status message is shown.

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
0.0003	Bob	Own inventory at the market	Sell 1,000 Ares limit \$100 GTC	Local access at Mars, sent h 0.0000	0.0003	Alice \$150k at Neptune; Bob \$50k at Mars, 3,000 Ares home
0.0003	Mars exchange	Its own available balances	Admits Bob EQ-sell; holds 1,000 Ares	—	—	Alice \$150k at Neptune; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Listing table: Ares trades at Mars	QUOTE for Ares	Local access at Neptune, sent h 0.0000	0.0003	Alice \$150k at Neptune; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Neptune exchange	No fresh Ares snapshot on its board	Quote request (MDREQ) for Ares to Mars	Backbone Ne→Ma, batched with queued records	12.3441	Alice \$150k at Neptune; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Alice	Own Neptune balance	Funds own Mars account: \$100k	Local access at Neptune, sent h 0.0000	0.0003	Alice \$150k at Neptune; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
0.0003	Neptune exchange	Instruction or fill; its own ledger	Debits Alice \$100k; TRANSFER NE-X1 (funding) to Mars	New session (SYN h 0.0006); DATA Ne→A h 8.2296, A→Ma h 12.0919	12.3441	Alice \$50k at Neptune, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
12.3441	Mars exchange	Its own live book	Replies with Ares snapshot (MDS, version 1)	Backbone Ma→Ne, batched with queued records	16.4594	Alice \$50k at Neptune, \$100k in transit; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
12.3441	Mars exchange	Authentic new transfer NE-X1	Imports once; credits Alice \$100k (spendable now)	RECEIVED to Neptune (backbone)	16.4594	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
16.4594	Neptune exchange	MDS from Mars	Posts Ares snapshot to its board (4.12 h old); answers waiting QUOTES	Local access (1 s)	16.4597	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
16.4594	Neptune exchange	Mars imported NE-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
16.4597	Alice	Home board (local access)	Reads Ares quote: bid none, ask \$100 × 1,000, 4.12 h old	—	—	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
16.4597	Alice	Published geometry; own funding confirmed	Sends ORDER ×8 copies	Direct Ne→Ma, launches 16.4597, 16.4763, 16.4930, 16.5097, 16.5263, 16.5430, 16.5597, 16.5763; p(loss) 0.906 each, all lost 4.6e-01	20.5653 (first)	Alice \$50k at Neptune, \$100k free at host; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
20.5653	Mars exchange	Its own available balances	Admits Alice EQ-buy; holds \$100k	—	—	Alice \$50k at Neptune, \$100k held; Bob \$50k at Mars, 2,000 Ares home, 1,000 Ares held
20.5653	Mars exchange	Both holds in its own ledger	Fill 1,000 Ares @ \$100: Alice buys from Bob	—	local, atomic	Alice \$50k at Neptune, 1,000 Ares at market; Bob \$150k at Mars, 2,000 Ares home
20.5653	Mars exchange	Instruction or fill; its own ledger	Debits Alice 1,000 Ares; TRANSFER MA-X1 (automatic home delivery) to Neptune	Session from h 0.0006; DATA Ma→A h 20.5656, A→Ne h 20.8186	24.6808	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.5820	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.5987	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6153	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6320	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6487	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
20.6653	Mars exchange	Same order already decided (ACCEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home

(continued)

h	Actor	Knows	Does	Packet or transmission	Arrives	Financial state after
20.6820	Mars exchange	Same order already decided (AC-CEPTED)	Ignores the copy; returns the original result	—	—	Alice \$50k at Neptune, 1,000 Ares in transit; Bob \$150k at Mars, 2,000 Ares home
24.6808	Neptune exchange	Authentic new transfer MA-X1	Imports once; credits Alice 1,000 Ares (spendable now)	RECEIVED to Mars (backbone)	28.7974	Alice \$50k at Neptune, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home
28.7974	Mars exchange	Neptune imported MA-X1	Marks transfer delivered; no balance change	—	—	Alice \$50k at Neptune, 1,000 Ares home; Bob \$150k at Mars, 2,000 Ares home

E4. Long-horizon scan (Tier 3)

Scan. We scanned 200 Julian years from the epoch at a 1-hour step (1,753,201 samples). At each sample we computed the clearance from the Sun, with a moving receiver, for all 38 directed backbone links and all 72 directed direct paths, without maintenance. For routes, we computed the best-route delay from every settlement to each market host and back at 6-hour steps, with flights in sequence.

Certification of the scan. The clearance is the Sun’s distance from the segment between the sender at t_e and the receiver at t_a . It changes no faster than the fastest endpoint moves plus the shift in arrival time, which gives a Lipschitz bound of $L \leq 0.00146$ AU per hour (Mercury at perihelion, 58.98 km/s). On any interval $[a, b]$ with end values $c(a)$ and $c(b)$, every interior value lies within $\frac{1}{2}(c(a) + c(b)) \pm \frac{1}{2}L(b - a)$. A second program (`e4_certify.py`) uses this as an interval test on all 110 paths. An hour is certified open if the lower bound is at least 0.10 AU and certified closed if the upper bound is below it. Otherwise it is split in half and both halves are tested again, down to 1 ms. Of the 192,852,000 path-hours, all but 116,956 were certified directly, and the refined hours contain 48,632 changes between open and closed, each located to within 1 ms. The closure counts agree exactly with the scan. Leaves that the test cannot classify can sit next to each other, so we merged adjacent uncertified leaves into connected intervals. There are 48,632 such intervals, exactly one per located transition. The longest is 0.29 s (337 leaves, at a grazing pass where the clearance stays very close to 0.10 AU), and every one has different states at its two ends, so each contains the change it is counted for. Any further changes inside an interval would come in pairs and bound a period shorter than the interval. Within the model, therefore, every change between open and closed in the 200 years is located to within 0.29 s, and no open or closed period longer than 0.29 s is missed. The run took 4.6 minutes.

Refined boundary (Tier 2). For the Earth to Relay A, first closure, the hourly samples bracket the boundary in $[24001200, 24004800]$ s. Bisection with the exact light-time solver narrows this to $[24004585.374355, 24004585.375214]$ s, a width of 0.858 ms, within the 1 ms tolerance. A packet emitted before the boundary follows its own path and is not affected. Our queue does not launch a packet after the boundary; the link waits instead. A closure therefore never destroys a packet in flight. It only delays the next launch, and the hop timer R_h restarts at the actual emission.

Results. The link between Relay A and Relay B never closes; its minimum clearance is 2.00 AU. Table A55 gives the results for the other links. Every settlement had a route to every host at every sample: the connected fraction is 1.000 for all 42 directed settlement–host pairs, as the geometric argument in the design paper predicts. The longest one-way route is Mars to Neptune at 315 minutes (hour 993,510). The longest direct closure is Jupiter to Saturn at 70.8 days. Direct paths are the weak point, and their closures are the reason for our access check, deferral and forwarding rule R8.

What holds beyond 200 years. Two results do not depend on the scan. The relay geometry argument shows that every settlement always has a route to every host. Since a route has at most three links, its length is at most $(r_a + R) + R\sqrt{2} + (r_b + R)$ AU, with aphelion distances r_a, r_b and relay radius $R = 2.828$ AU, so

Table A55: Results of the 200-year scan. Relay columns: availability, number of closures and longest closure. Host columns: one-way route time in minutes (minimum / median / maximum). Last column: availability and longest closure of the direct path to Mars.

From	To Relay A	To Relay B	To Mars	To Ceres	To Earth	Direct to Mars
Mercury	90.9%, 788, 267 h	90.9%, 789, 267 h	30/42/86	20/43/91	35/43/84	89.7%, 14 d
Venus	94.6%, 283, 345 h	94.6%, 283, 345 h	27/43/87	19/43/91	33/44/85	93.6%, 24 d
Earth	95.7%, 158, 493 h	95.7%, 158, 493 h	25/43/87	17/43/92	local	94.7%, 53 d
Mars	96.9%, 64, 1086 h	96.9%, 64, 1084 h	local	12/44/94	25/43/87	local
Ceres	99.5%, 2, 4565 h	99.7%, 2, 3768 h	12/44/94	local	17/43/92	98.9%, 38 d
Jupiter	98.3%, 25, 1243 h	98.4%, 25, 1211 h	30/59/112	23/65/106	33/58/110	97.4%, 26 d
Saturn	99.0%, 35, 747 h	99.0%, 36, 749 h	62/95/148	53/99/144	67/92/145	97.8%, 20 d
Uranus	98.7%, 40, 571 h	98.7%, 40, 571 h	139/174/227	134/178/230	145/172/227	97.8%, 17 d
Neptune	99.0%, 41, 528 h	99.0%, 41, 529 h	235/263/315	227/268/309	240/261/312	98.0%, 17 d

every one-way route delay with empty queues is bounded for all time (`e4_bound.py`). The largest bound is 358 minutes (Neptune to Ceres, against 309 observed), and every observed maximum is at least 9 minutes below its bound. The availability figures, closure counts and typical delays in Table A55 are statistics of the scanned 200 years only. The orbital periods are not commensurate, so the configuration never repeats exactly, and we do not extrapolate them. Section E5 tests three later epochs directly.

E5. Shifted epochs (Tier 3)

Each run starts at $t = k \times 365.25$ days with every orbit advanced from the epoch. Balances are reset to the opening values, maintenance and the S2 incident are not repeated, and the observation and maturity times move with the start. S1d and S1e give identical times at every epoch (only the payout split differs), so Table A56 lists them together.

Table A56: Results at shifted epochs. Route columns: backbone route used and one-way time in minutes, then the direct path time and 24-hour availability (backbone availability is 100% unless shown). Scenario columns: completion time without loss, with the median of 300 lossy runs in brackets. Where the result without rule R8 (forwarding) differs, it is shown after “no R8”.

Epoch	Start (h)	Earth–Mars	Ceres–Mars	S1a equity	S1d/S1e futures
+1 year	8,766	Ea-B-Ma 47; direct 17, 100%	Ce-B-Ma 32; direct 23, 100%	4.22 (5.76)	329.87 (331.91)
+10 years	87,660	Ea-B-Ma 42; direct 22, 0%	Ce-B-Ma 19; direct 11, 100%	3.87 (3.89); no R8: 350.75	329.27 (331.32); no R8: suspended
+100 years	876,600	Ea-B-Ma 39; direct 18, 100%	Ce-A-Ma 50; direct 37, 100%	3.52 (3.53)	330.48 (332.37)
Mars conjunction	12,470	Ea-A-Ma 50; direct 20, 0%	Ce-B-Ma 42; direct 31, 100%	4.55 (6.06); no R8: deferred	330.23 (332.79); no R8: suspended

Difficult epoch. We chose the first Mars solar conjunction after the epoch. The Earth to Mars direct path closes at hour 12,446.0 and reopens at hour 13,713, 53 days later, and we start 24 hours after the closure. With rule R8, Alice’s client forwards her instructions through Mercury, whose direct legs to Earth and to Mars are both open. Her order goes Earth to Mercury as 2 copies (each lost with probability 0.069), and Mercury forwards it once as 3 copies (each lost with probability 0.125). The first copy reaches Mars at hour 3.708, with probability 0.814 that both first copies arrive. The probability that some copy crosses each leg, so that the order arrives within minutes (by about hour 3.76, if only the last first-leg copy gets through), is 0.9933. Mercury uses at most 3 of its 12 daily direct packets. Alice learns that her contract is OPEN at hour 4.55 from a STATUS record over the backbone. The equity trade completes in 4.55 hours, and both futures price paths complete in 330.23 hours. With random loss, 300/300 equity runs (median 6.06 h, p99 11.03 h) and 300/300 futures runs (median 332.79 h) complete. A lost forwarded copy is recovered by the R6 fallback as attempt $k + 1$.

Without R8, Alice’s client finds no long enough direct window and never funds Mars: no trade takes place, and Cara’s futures margin is back home by hour 170.22. Nobody loses anything, but nobody trades. At +10 years the equity trade waits 351 hours and the futures are suspended.

Which path each instruction took. Table A57 lists every direct leg in the shifted-epoch runs of S1a and S1d. At +1 year and +100 years every instruction goes direct. R8 forwards instructions at +10 years and Mars conjunction, through the exchange named in the Sender column of the second leg.

Table A57: Every direct leg in the E5 runs of S1a and S1d (no loss). A forwarded instruction has two legs: the client to the forwarding exchange and that exchange to the host; it arrives with the product of the two P(arrives). Leg: fwd 1 is client to forwarder, fwd 2 is forwarder to host.

Epoch	Run	Sender	Path	Leg	Message	Copies	p	each	First	h	P(arrives)
+1 year	S1a	Alice	Ea→Ma	direct	ORDER	3	0.151	3.43	0.9966		
+1 year	S1d	Cara	Ce→Ma	direct	CINSTR	3	0.196	2.51	0.9925		
+1 year	S1d	Alice	Ea→Ma	direct	CINSTR	3	0.151	3.43	0.9966		
+10 years	S1a	Alice	Ea→Me	fwd 1	ORDER	2	0.067	2.92	0.9956		
+10 years	S1a	Mercury (R8)	Me→Ma	fwd 2	ORDER	3	0.137	3.17	0.9974		
+10 years	S1d	Cara	Ce→Ma	direct	CINSTR	3	0.102	1.44	0.9989		
+10 years	S1d	Alice	Ea→Me	fwd 1	CINSTR	2	0.067	2.92	0.9956		
+10 years	S1d	Mercury (R8)	Me→Ma	fwd 2	CINSTR	3	0.137	3.17	0.9974		
+100 years	S1a	Alice	Ea→Ma	direct	ORDER	3	0.162	2.88	0.9957		
+100 years	S1d	Alice	Ea→Ma	direct	CINSTR	3	0.162	2.88	0.9957		
+100 years	S1d	Cara	Ce→Ma	direct	CINSTR	4	0.296	3.94	0.9923		
Mars conjunction	S1a	Alice	Ea→Me	fwd 1	ORDER	2	0.069	3.47	0.9953		
Mars conjunction	S1a	Mercury (R8)	Me→Ma	fwd 2	ORDER	3	0.125	3.71	0.9980		
Mars conjunction	S1d	Cara	Ce→Ma	direct	CINSTR	4	0.258	3.29	0.9956		
Mars conjunction	S1d	Alice	Ea→Me	fwd 1	CINSTR	2	0.069	3.47	0.9953		
Mars conjunction	S1d	Mercury (R8)	Me→Ma	fwd 2	CINSTR	3	0.125	3.71	0.9980		

E6. Quota, backlog and storage

Two quota classes. Each exchange may originate 66 backbone packets in any rolling 24 hours. A data packet that carries any record other than RECEIVED is *routine* and may be sent only while the exchange has made fewer than 44 originations in the last 24 hours. SYN packets, packets that carry only RECEIVED records, and application resubmissions may use all 66. So at least 22 originations a day always remain for recovery and receipts, however much new work is queued. The simulator enforces both limits (`transport.Quota, reserve=22`). When routine work is blocked, waiting receipts are taken out of the queue and sent in packets of their own.

Backlog experiment. `backlog.py` puts a queue of \$100 transfers into the outboxes at hour 0, which is the situation at the end of a long outage, and runs until everything is resolved (Table A58). Three hundred transfers to one peer fill 60 data packets of five, plus one SYN. Earth sends 44 originations at once and the remaining 17 when the first ones leave the 24-hour window, so all transfers are imported by hour 24.6 and resolved by hour 25.1. Mars uses 60 originations for the receipts, which it may send from its reserve. With random loss every run cleared, with a median of 26.8 hours. Traffic in both directions costs little extra, because receipts share packets with the reverse transfers. A backlog to all eight peers at once is limited by Earth’s own 44 routine originations a day: 2,400 transfers need about 490 packets, or 11 days, and they resolve by hour 248. The 1,200-transfer case shows the window: the first 512 transfers go out and the rest wait until receipts come back.

What “clears” means. A backlog has cleared when every transfer is credited at its destination and the source has its RECEIVED, so no unresolved state remains. Credit comes earlier than resolution by at most one route delay plus the receipt’s queueing. The claim in the design paper is therefore: *when no other traffic competes, 300 transfers to one peer are credited within about 25 hours and resolved within about 26 hours of the route reopening; competing traffic from the same exchange shares its 44 routine originations a day.*

Table A58: Backlog after an outage. All transfers are already in the outbox at hour 0. One peer: 300 transfers from Earth to Mars. Both ways: 300 each way between Earth and Mars. All peers: 300 from Earth to each of the 8 other exchanges. Over window: 1,200 from Earth to Mars. “All imported” is when the last one is credited at its destination and “All resolved” when the source holds every RECEIVED (hours, no loss). N is the number of transfers. Orig. and Peak are Earth’s originations in total and in its busiest 24 hours (limit 66, of which 44 routine). The lossy columns use 20 runs (5 for the last row): runs fully resolved within 40 days, median and maximum time to resolve, and (Peak any) the highest 24-hour origination count of any exchange in any run. Held counts transfers that waited for the window of 512.

Case	N	Orig.	Peak	Imported	Resolved	Cleared	Median	Max	Peak any	Held
One peer	300	61	44	24.6	25.1	20/20	26.8	30.0	52	0
Both ways	600	109	65	24.6	25.7	20/20	28.3	31.5	65	0
All peers	2,400	490	51	244.2	248.4	20/20	330.5	445.0	66	0
Over window	1,200	241	44	120.6	121.1	5/5	124.3	127.6	66	688

Window and compaction. A source numbers its transfers to each destination consecutively, sends them in that order, and does not send transfer n while any transfer numbered $n - 512$ or lower is unresolved (equivalently, $n < u + 512$ for the lowest unresolved number u). Let u be the lowest unresolved number at the source. Every transfer below u has a RECEIVED, so it was credited and the destination’s watermark satisfies $W \geq u - 1$. The source has sent nothing at or above $u + 512$. The credited transfers above W therefore all lie in $[u, u + 511]$, so the destination never keeps more than 512 tombstones per source, about 23 KB, however long a gap stays open. A transfer held by the window has already been debited and is durable at the source; it is in transit and still owned by the client, exactly like a transfer waiting for an open link. The simulator implements the window (`protocol.XFER_WINDOW`); in the 1,200-transfer case 688 transfers were held and released in order.

Admission and reservation. An exchange has 4,096 export slots. A slot is in use while an export is unresolved or while it is *reserved* for a transfer the exchange has promised. Accepting a remote client’s contract instruction reserves one slot for that party’s payout home; the slot is freed if the contract never opens, and used by the payout otherwise. Accepting a remote client’s order with home delivery reserves one slot for its delivery. An order has at most one delivery in transit: shares filled while it travels are added to a pending amount, sent in one transfer when the RECEIVED returns, and the slot is kept while the order rests. Funding, withdrawal and close each need slots for their own transfers. An instruction is refused (“try later”) if the slots in use plus those it needs would exceed 3,276; local trading continues. A payout or delivery therefore always finds its slot already held. The live-order limits (32 per client, 256 per host) are enforced at order entry.

Returns of late top-ups. A transfer to a closed account is the one export the host cannot refuse in advance: the home may have sent it before it learned of the closure, and each of the eight other exchanges may have up to 512 transfers in flight, so their number is not bounded by the spare 820 slots. The host credits such a transfer to a *restricted* balance, still owned by the client and usable for nothing but its return, and records one return entry per closed account, asset and source; a later top-up for the same entry adds to it. A return is exported only while slots in use are below the hard cap of 4,096, oldest entry first; otherwise it waits. Because returns use only free slots and admission counts them, slots in use never exceed the cap, and the reserved slots of payouts and deliveries are never taken. The waiting entries are bounded by the number of closed accounts times assets times sources, each one balance row. Every waiting return keeps a funded owner: the money stays in the client’s restricted balance at the host until the return is exported, and is then in transit, still the client’s, until it is credited at home.

Synthetic capacity tests. `saturation.py` tests the ledger’s capacity rules, not the network scenario: it is not a compliant workload under the brief. It uses about 5,250 synthetic client identities instead of the six opening accounts, injects their instructions at the Mars ledger without the 12-packet direct quota or R6 retries (2,000 contract instructions alone would need at least 667 direct packets, against 72 a day for the six real clients), and places their cash at Mars directly instead of funding it. Backbone transport, its quotas and the transfer window are real. Within two hours Mars receives 2,000 contract instructions (1,000 contracts of 48 hours), 250 home-delivery buy orders filled three shares every 30 minutes, and 3,000 withdrawals. Table A59 compares the reservation rule with the earlier rule that counted only unresolved exports. Under the earlier rule every request was accepted and unresolved exports reached 4,986, above the cap for 123

hours: the payouts were owed but had no capacity. Under the reservation rule the withdrawals that did not fit were refused at the door, slots in use never exceeded 3,276, and every payout and delivery went out.

Table A59: Synthetic capacity test at Mars (not a compliant network scenario): the reservation rule against the earlier admission rule.

Mars at its limits	Reservation	Unresolved only
Contract instructions accepted / refused	2,000 / 0	2,000 / 0
Delivery orders accepted / refused	250 / 0	250 / 0
Withdrawals accepted / refused	1,026 / 1,974	3,000 / 0
Payouts resolved	2,000 of 2,000	2,000 of 2,000
Shares delivered (transfers)	2,500 of 2,500 (500)	2,500 of 2,500 (500)
Peak unresolved exports	3,012	4,986
Peak slots in use (incl. reserved)	3,276	5,250
Hours above the 4,096 cap	0.0	122.5
Last RECEIVED (h)	560	800
Pending shares / reserved slots at end	0 / 0	0 / 0

A second test fills Mars to the admission limit with 3,276 withdrawals in flight and then delivers 2,400 top-ups from all 8 other exchanges (300 each) to accounts already closed at Mars, three times the spare slots. Slots in use peaked at 4,096 and never exceeded the cap; up to 1,419 returns waited at once. All 2,400 returns were resolved, the last at hour 745; every client's \$1,000 was back at home, and no restricted balance remained.

Table A60 lists every bounded part of an exchange's state.

Retention of request results. Withdrawal, funding, cancel and close requests carry a per-account request number and a creation time. The exchange keeps each result (request ID, outcome, terms digest and transfer ID, 57 bytes) for 60 days after the creation time, which covers the brief's 30-day packet lifetime and its 30-day duplicate retention. A copy created more than 30 days ago is rejected on that ground alone, so a dropped result can never be executed twice. A closed account keeps a 52-byte closure record so that a late top-up is returned, at least until every return for it is resolved. Once the home exchange learns of the closure it accepts no more funding for that account, so the record is dropped when the host's watermark for the home passes the last transfer the home sent before learning of it. A contract record becomes a tombstone once both payouts are resolved and is dropped 30 days after the opening deadline, when every copy of its instructions has expired. These retention rules are design rules: the simulator keeps every result and record for the whole run (at most 1,600 hours), so it does not exercise the 60-day expiry.

E7. Market data

Design. Every exchange holds the listing table and a board with the newest snapshot of each stock (R5). A client reads both by local access, at no quota cost; at the market itself the board is the live book. Snapshots travel only as the reply to a quote request: a client's QUOTE instruction makes its exchange ask the market over the backbone if the board is older than the client accepts, at most once per stock per 6 hours. In S1 the client sends QUOTE together with its funding, so the request shares the funding transfer's packet and the reply shares the RECEIVED's. Packet counts, originations and completion times are identical to runs without market data, in S1 and in 60 lossy S1a, S1c, S1d and S1i runs, and the client reads the quote just before it orders (mark `quote_seen`).

Load test. `md_test.py` has a client at every settlement ask for every stock listed elsewhere (24 exchange-stock pairs) for 30 days, with random loss and maintenance, while a local trader at each market changes the book every 2 hours. Table A61 gives the waits and ages. With a request every 6 hours, which is the most an exchange sends, each market sent 14.4 replies a day and its busiest 24 hours used 23 originations including session setup. With one request a day it sent 4.0 and used at most 12. 47% of the 6-hourly requests were answered from the board at once. Waits are set by distance: a request and its reply each cross the backbone, and at Uranus and Neptune a new session's handshake and losses of about a third per hop add a day or more. The quote never gates safety: an order's limit price bounds what it pays.

Table A60: Bounded state at each exchange.

State	Limit	When the limit is reached	Note
Live orders	32 per client per host; 256 per host	Order refused	Orders expire within 30 days
Export slots (unresolved exports plus reserved payouts and deliveries)	4,096 per host	An instruction needing slots is refused if slots in use would exceed 3,276 (80\%)	Payouts and deliveries hold their slot from acceptance; returns of late top-ups use only free slots and otherwise wait
Late top-ups awaiting return	One restricted balance per closed account, asset and source	Waits, owned by the client, until a slot is free	Later top-ups merge into it; never more slots than the cap
Transfers in flight to one destination	Window of 512	Transfer n waits (debited, durable) while any transfer numbered n – 512 or lower is unresolved	Sent in number order; bounds the destination’s tombstones
Transfer tombstones above the watermark	At most 512 per source	Cannot be exceeded	45 bytes each: ID 12, state 1, terms digest 32
Request results (withdrawal, funding, cancel, close)	Kept 60 days after the request’s creation time	A copy older than 30 days is rejected as expired	At most 64 requests per account per host per day
Account closure record	52 bytes until the host’s watermark for the home passes the home’s last transfer sent before it learned of the closure, and every return is resolved	—	Diverts late top-ups; the home sends none after (not simulated)
Contract records	Full record until both payouts are resolved, then a tombstone until 30 days after the opening deadline	—	Every instruction copy has expired by then (not simulated)

Table A61: Quote requests by clients at each exchange, pooled over the stocks listed elsewhere, in 10 lossy 30-day runs per demand level. Wait: hours from the QUOTE instruction to the quote (0 when the board was already fresh enough). Age: hours since the market took the snapshot, when the client received it. Daily: one request per stock per day, accepting quotes up to 24 hours old. 6 h: one every 6 hours, accepting quotes up to 6 hours old.

Exchange	Daily wait 95%	Daily age median	6 h wait 95%	6 h wait max	6 h age median	6 h age 95%
Mercury	3.3	19.7	3.2	8.0	2.7	5.3
Venus	2.8	16.9	2.8	7.8	2.3	5.4
Earth	2.7	19.2	2.6	5.8	2.4	5.5
Mars	2.6	20.7	2.6	4.3	2.3	5.5
Ceres	2.6	19.1	2.5	5.5	2.4	5.5
Jupiter	4.4	13.1	3.6	9.8	2.8	5.4
Saturn	5.7	12.7	5.7	29.3	3.2	4.7
Uranus	18.0	10.4	19.7	51.2	3.4	13.8
Neptune	43.4	12.9	49.4	84.5	5.1	30.4